Veritas Document 3A

FEAT Fairness Principles Assessment Methodology

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Executive Summary

As a concept, fairness is not novel to decision making, and certainly not unique to Artificial Intelligence & Data Analytics driven decision making ("AIDA", which is defined by the Monetary Authority of Singapore as "technologies that assist or replace human decision-making"). In a general sense, fairness involves seeking what is right, impartial and equal without favouritism or discrimination.

AIDA systems powered by high quality data, sophisticated algorithms and ever-increasing processing capabilities, bring scale and value to data driven decision making processes. However, without careful design and control, these systems can bring new risks and unintended harms, perpetuating or reinforcing existing disadvantages in society or even introducing new ones. With so many decisions in financial services aided by AIDA systems, the fairness of these systems is becoming increasingly important.

More and more, regulators worldwide are playing a critical role in guiding Financial Service Institutions (FSIs) to prevent harms and deliver benefits. In Singapore, within the Fairness, Ethics, Accountability and Transparency (FEAT) Principles, the Principles that explicitly relate to fairness (see figure P1-4 in section 1.1.1) aim to ensure that AIDA driven decisions do not systematically disadvantage individuals or groups of individuals, without appropriate justification.

To help organisations apply the Principles, Veritas Phase 1 (published in January 2021) formulated the initial FEAT Fairness Principles Assessment Methodology ("Fairness Methodology"). In Phase 1, use case examples of the "Fairness Methodology" focused on the banking sector. This document is the output of Veritas Phase 2 and it extends the Phase 1 Fairness Methodology with frameworks, practical guidance and examples, covering both banking and insurance. The Fairness Methodology provides a guide for FSIs to work towards assessing alignment of their AIDA systems' operations with the Fairness Principles.

The main extensions to the Fairness Methodology in Phase 2 are:

- Key concepts such as personal attributes, types of bias and mitigation methods, and fairness objectives and metrics are defined in more detail, along with illustrative frameworks and guidance on how to identify and apply them to a use case.
- Parts A-E of the Fairness Methodology defined in Phase 1 are aligned to a typical AIDA system development lifecycle, enabling FSIs to develop fair AIDA applications by design.
- Detailed considerations and guidance are included for each of the 18 Fairness Methodology assessment questions from Phase 1 to help FSIs think through the important aspects to consider when answering the questions.
- The 18 Fairness Methodology assessment questions from Phase 1 are integrated into the overall FEAT checklist to enable a full FEAT assessment of the end-to-end AIDA system.
- Guidance is provided on scaling and embedding the Fairness Methodology across an FSI, through
  - Defining the internal foundational set of practices and standards for AIDA fairness,
  - Including guidance on how to customise the depth of the fairness assessment appropriate for a given AIDA system, and
  - Including guidance on how FSIs can integrate AIDA fairness into the fabric of their organisation.

The use case example in Phase 2 is predictive underwriting for life insurance. A simplified underwriting journey for existing customers of an insurer in Singapore and a detailed illustration of the use case can be found in the Phase 2 Illustrative Case Studies document.

01 Introduction

1.1 Purpose and Intended Audience of the Document

Accelerating digitalisation is affecting Financial Services Institutions (FSIs) through two overlapping waves – a behavioural shift and technological shift. Behaviourally, consumers are increasingly looking for FSIs to provide them with digital services, and for these services to be personalised to their needs. Technologically, a broad range of innovations are impacting on the digital transformation of FSIs, including digital platforms and marketplaces and advanced sensor and analytics capabilities. These technological developments are being powered by the rise of agile, digitally native entrants as well as a substantial increase in computational power.

The adoption of Artificial Intelligence and Data Analytics (AIDA) systems is helping FSIs take advantage of this evolving environment to better support decision making through superior insights from new and existing data. Additionally, FSIs are leveraging AIDA systems to increase productivity, both through the data driven automation of existing processes and by helping them to develop wholly new ways of doing business. By supporting growth, delivering better customer experiences and reducing the risk and cost of doing business, FSIs are looking to scale through use of AIDA systems and stay future ready and relevant.

In turn, strong data driven approaches bring benefits to society through the creation of innovative products and services that support affordability, ease of access and the quality of the end service, while also helping to bridge financial inclusion and protection gaps. Additionally, with better access to data and the availability of digital touchpoints across the service value chain, FSIs can increase the accuracy of their decision making to drive better outcomes for customers. By adopting AIDA systems at scale, FSIs hope to improve insights, predictive analytics and service personalisation to accelerate differentiation, while also creating more efficient processes and increasing the value of the tasks undertaken by employees.

As FSIs deliver against their data strategies by upgrading their decision making process and associated customer services, it is vital to ensure that AIDA enabled decisions, which potentially touch millions of lives, do not reflect unintentional discriminatory behaviour.

AIDA systems could also introduce new risks for businesses and customers that need to be properly managed and mitigated. Poorly designed AIDA systems can cause unintended harms, breach applicable laws, rules and regulations and perpetuate or reinforce existing disadvantages in society. These harms can arise due to the fact AIDA systems operate according only to the data and objectives explicitly provided by their designers, without taking into consideration the broader context or moral constraints that humans often take for granted. For example, when an AIDA system is designed to be accurate for the overall population this can be achieved unintentionally in a way that leads to a disproportionate number of erroneous decisions being made on disadvantaged minority populations because they are poorly represented in the data. Systematic erroneous decision making by AIDA systems can also damage the reputation of an FSI and affect its business.[2]

There is therefore a clear need for fairness to be integrated throughout the AIDA delivery cycle phases while bringing AIDA systems to scale.[3]

This document is one of a suite published as a ¬deliverable of the Monetary Authority of Singapore (MAS) Veritas Phase 2 project. Its purpose is to define a Fairness Methodology for FSIs bringing AIDA systems to scale, and it fits alongside published documents as highlighted in Figure 1.1.

FSIs should continue to comply with all applicable laws and requirements. The Fairness Methodology is aspirational, promoting fairness of AIDA systems and not intended to replace existing relevant internal governance frameworks. FSIs are encouraged to calibrate their internal governance frameworks for the FEAT fairness assessment based on their own discretion taking into consideration their existing frameworks, materiality of AIDA systems and cost of FEAT fairness assessments and potential mitigation.

The illustrations used in this whitepaper are only applicable in the context of the jurisdiction of Singapore, and only for the specific attributes assessed – they are not generalisable.

The Fairness Methodology described in this whitepaper is only applicable if "personal attributes", including personal and sensitive personal data, are collected and processed in accordance with all applicable regulatory requirements. Personal attributes are defined in section 2.1.4.

"Personal data" is defined as data, whether true or not, about an individual who can be identified either from that data or from that data and other information to which the organisation has or is likely to have access.

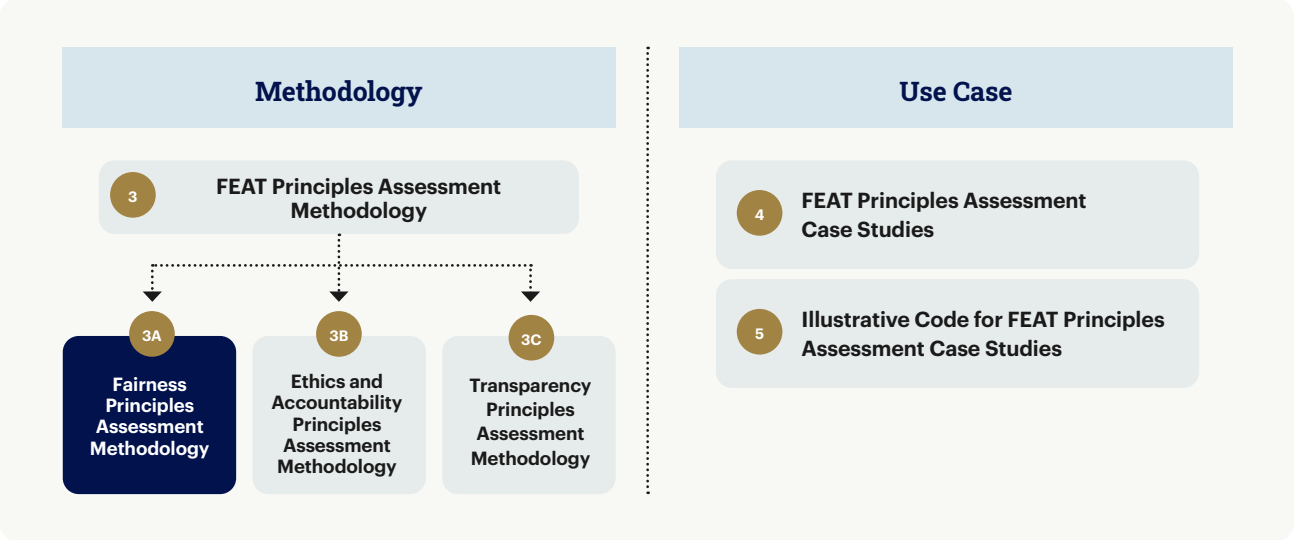


Figure 1.1 Fairness Principles Assessment Methodology document in the suite of Veritas Phase 2 outcomes

Please refer to section 1.3 of Veritas Document 3 FEAT Principles Assessment Methodology document to review the intended audiences for this document.

1.1.1 Background

The Principles of Fairness ("Principles"), under the Monetary Authority of Singapore's FEAT Principles[4], published in 2018, focus on two key aspects - "justifiability" and "accuracy and bias":

Justifiability

P1 Individuals or groups of individuals are not systematically disadvantaged through AIDA driven decisions, unless these decisions can be justified.

P2 Use of personal attributes as input factors for AIDA driven decisions is justified.

Accuracy and Bias

P3 Data and models used for AIDA driven decisions are regularly reviewed and validated for accuracy and relevance, and to minimise unintentional bias.

P4 AIDA driven decisions are regularly reviewed so that models behave as designed and intended.

Figure 1.2 MAS FEAT Fairness Principles (2018)

To move the Principles to practical application, Veritas **Phase 1** (published in January 2021) formulated the initial FEAT Fairness Principles Assessment Methodology ("Fairness Methodology"). Use case examples of the "Fairness Methodology" focused on the **banking sector** and applied to credit scoring and customer marketing AIDA systems were provided in Phase 1.

1.1.2 Scope

This document is the output of Veritas **Phase 2**, which commenced in March 2021. **It extends the Phase 1 Fairness Methodology** with illustrative frameworks, practical guidance and examples covering **both banking and insurance in the context of individual data subjects.** The Fairness Methodology provides a guide for FSIs to assess alignment of their AIDA systems' operations with the Principles.

The Principles ask FSIs to consider both whether the AIDA driven decision making process and the outcomes from these decisions are justified. Moving Principles to practical application requires FSIs to determine and define fairness objectives for various applications of AIDA, outline specifications to measure fairness and account for these in the AIDA systems, as well as in existing governance processes.

The document acknowledges that an FSI's specific obligations around AIDA fairness will depend on the regulatory requirements, societal and legislative norms or case law of the jurisdiction in which it operates, as well as the organisation's values and existing governance standards, which are also likely to change over time. Therefore, the Fairness Methodology is not prescriptive, but instead provides guidance and recommendations for relevant audiences. **It Is important that the level of FEAT fairness assessment is proportional to the fairness risk of any use case as these assessments will need to be implemented with additional costs, which ultimately get passed on to consumers.**

Also, it should be recognised that the area of algorithmic fairness is still developing and for some complex areas, there is ongoing active research to determine how to address current challenges.

The document covers the following themes:

- Section 1 outlines what fairness is and why it is important.
- Section 2 introduces the reader to key concepts in the fairness principles
- Section 3 details the Fairness Methodology, which is designed to aid FSIs in making their fairness objectives explicit and for providing evidence that their AIDA systems' operations meet those objectives.
- Section 4 provides guidance on how to embed and scale the Fairness Methodology across an FSI and on how to customise the application of the Fairness Methodology for the fairness risk level of the AIDA system in question in a scalable way.
- Section 5 outlines fairness areas for future considerations by the industry and regulators

1.2 The Importance of Fairness in AIDA Driven Decision Making

1.2.1 What is Fairness?

The idea of fairness is used in various disciplines including philosophy, sociology, law and politics. However, how fairness operates or what is considered fair remains highly contested.[5]

Fairness is closely related to human rights, non-discrimination, solidarity and justice.[6] These ideas revolve around things that are right, impartial, equal, and delivered without favouritism or discrimination as assessed by evaluating the consequences of a decision, along with the process followed to reach the decision.[7]

There are many notions of fairness, including basing this on equality or need.[8] Notions of fairness continue to be debated in both moral philosophy and wider society. As one notion of fairness may conflict with another, fairness is an "essentially contested" concept.[9] Therefore, although the general nature and importance of fairness are widely understood, its precise definition and what constitutes fair or unfair outcomes, should be defined in a given context (i.e., fairness is use-case specific).

According to the Guidelines published by the European Commission's High-Level Expert Group (HLEG) on Artificial Intelligence,[10] fairness has two dimensions:

- **The substantive dimension.** Ensuring the equal and just distribution of benefits and costs and ensuring that individuals and groups are free from unfair bias, discrimination and stigmatisation.
- **The procedural dimension.** This includes seeking effective redress against decisions made by AIDA systems (and the humans managing/operating them), with an effective governance framework and controls in place.

The substantive dimension is within the scope of this whitepaper, while the procedural dimension is covered in the Ethics and Accountability whitepaper.

1.2.2 (Why) Do AIDA Driven Decisions Require Enhanced Fairness Controls?

As a concept, fairness is not novel to decision making and certainly not unique to AIDA driven decision making. When humans act or decide, they are held responsible for the accuracy, reliability, and soundness of their judgements. Moreover, it is often demanded of them that their actions and decisions are supported by good reasons, and they are held accountable for the justness, equity, and reasonableness of how they treat others.

However, as a technology, AIDA systems are not accountable agents. Accountability remains with the people and organisations who choose to deploy them.[11] It is important that there is clear responsibility — **defined by FSIs** — for the ownership of AIDA driven decisions and the soundness of an AIDA fairness assessment for **reasonably foreseeable risks**.



AIDA systems bring value to data driven decision making processes. However, without careful design and control, these systems can bring new risks and unintended harms, perpetuating or reinforcing existing disadvantages or even introducing new ones.[12] In some cases, these harms are an outcome of FSIs' decisions on the design and development of their AIDA systems. For example, if an AIDA system is simply designed to optimise overall accuracy, the distribution of erroneous decisions may not be equitable across subpopulation groups. The following key factors have led to increased attention on fairness (and ethics, in general) of AIDA driven decisions:

- **Ability of automated AIDA decisions to scale and impact a wider group of customers.** AIDA driven decisions can impact more people and be perpetuated at greater speed than manual or traditional automated alternatives. Erroneous decisions can therefore harm a larger number of people. In cases of unintended outcomes or poor decisions, the associated legal or reputational damage for the FSIs can therefore also be much greater than in the past.
- **Ability of AIDA to differentiate to a finer degree than historical human led approaches by seeing patterns in large datasets.** AIDA may learn from latent relationships in the data (even when personal factors are obscured) or from historic biases and may differentiate outcomes in unknown (and unintended) ways.
- **Use of novel sources of data to predict outcomes.** There is an increasing use of novel sources of consented personal data by FSIs. In insurance, for example, the use of electronic health records, lifestyle markers and telematics can drive value (e.g., personalised engagement, better prediction, better prevention, better risk management, etc.). However, the use of such data could lead to concerns around the credibility, integrity and reliability of third party data sources and associated privacy implications. Another challenge is around how to verify an identity across multiple, independent datasets without breaching the privacy of the individual concerned (i.e., the data subject). Knowledgeable customers now seek robust justifications when their data is used to drive decisions.
- **Involvement of third party developers.** A lack of visibility into the delivery lifecycle process and the fairness assessments conducted by third party organisations makes it harder for FSIs to ensure fairness when using externally developed AIDA systems.

While best practice, frameworks, laws and regulations in this area are evolving, it is nonetheless the responsibility of FSIs to identify and address unfair outcomes from their AIDA systems to ensure that AIDA driven decisions do not systematically disadvantage individual or groups of individuals without justification. To do so, systems should be designed with **explicit and quantifiable fairness objectives**.

Since one notion of fairness may conflict with others and differ by application use case, it is not feasible to attempt to build a system that is universally recognised as fair. Instead, FSIs will need to make decisions on trade-offs between opposing fairness metrics, as well as potentially between their fairness and and commercial objectives.

Fairness assessments should be carried out on AIDA systems rather than the models alone. The assessment should not be confined to the mathematical properties of models. Rather, it should include a holistic assessment of risks and factors contributing to the AIDA driven decisions (such as business rules and human oversight) and the resulting outcomes for the individuals affected. The fairness assessment and review should be performed throughout the AIDA system development lifecycle, including monitoring on the live AIDA system. The depth of the assessment should be proportionate to the fairness risk that the AIDA system in question poses (see section 4.1 for guidance on how to assess the fairness risk of an AIDA system).

1.2.3 Fairness in AIDA: Regulatory Context

As FSIs drive the development and adoption of AIDA to unlock efficiencies and business value, regulators worldwide are stepping up to help guide FSIs to prevent unintended harms and deliver benefits across society.

There are various frameworks and policies concerning AIDA globally, and a shift from non-binding principles, guidelines and policy papers to legally binding laws and regulations is to be expected. Examples of non-binding AIDA principles is available in appendix 8.12.

It is important to note that fairness is also a data protection and privacy principle. Where algorithmic processing involves personal data, FSIs have a legal requirement to comply with Singapore's Personal Data Protection Act (PDPA), which focuses on the underlying reasons and legal basis that justifies the collection, processing and use of customers' personal data by an FSI.

FSIs' data and model governance rules should ensure that protected factors are not used in the decision making process and that there is a regular model review process which considers relevant legislative changes.

The nature of personal data protected by law can vary between industries. For example, the US Equal Credit Opportunity Act (ECOA) specifically targets industries that provide credit services and specifies that gender must not to be used as a factor when deciding on the outcome of a credit application. Accepted demographic attributes are specific to each respective use case, and it is important to recognise that what is considered appropriate demographic attributes can vary by jurisdiction. Moreover, some jurisdictions incorporate employee data protection and anti-discriminatory laws in the field of employment and labour laws. Regulations around anti-discrimination standards are constantly evolving as new trends emerge.

Concurrently, privacy concerns are arising as companies feed increasing volumes of consumer and vendor data into advanced AI algorithms as part of their business operations. Consumers are paying more attention to their private information and becoming increasingly interested in how their data is used by the entities with which they interact.[13] Giving individuals genuine choice in providing consent for the gathering and use of their personal data, and securing and managing this consent, should be the standard operating procedure for every FSI.

In the EU, the consent requirement for the processing of particularly sensitive personal data as protected under Article 9 of the General Data Protection Regulation (GDPR) is intended to avoid discrimination against individuals on the basis of personal characteristics. Article 9 sets in stone Article 21 of the Charter of Fundamental Rights of the EU (non-discrimination). To ensure that consent is obtained in a fair manner, it is important that data subjects can make an informed decision. FSIs need to give customers control over how their data is used, and the onus lies with FSIs to ensure the appropriateness of the data they collect and use, in line with customer consent and data privacy requirements.

According to data protection and privacy laws, to ensure the fair and transparent processing of data in respect to the data subject, FSIs should consider the specific circumstances and context in which the personal data is processed. For example, the EU's GDPR requires FSIs to use appropriate mathematical or statistical procedures for profiling. Significantly, GDPR only allows automated decision making and profiling based on special categories of personal data under specific conditions.

One of fairness considerations related to data protection and privacy in the regulatory context is how FSIs adhere to applicable antitrust and competition laws, factors that affect pricing and consumer welfare. The rise of data gathering at scale, and the ability to process data through AIDA systems is very often discussed in light of competition law. The discussion often focuses on whether the algorithms used by an AIDA system increase the potential risk of collusion between competitors, in particular when it comes to pricing algorithms and providing access to AIDA system on a non-discriminatory basis.[14]

FSIs therefore need to monitor evolving regulatory standards around anti-discrimination, both locally and globally, along with respecting consumer consent and data privacy rules. This is to ensure that the attributes identified for use in decision making through AIDA systems and fairness assessments are justified for use and prevent unintended harms resulting from the systems.

02 Key Fairness Concepts

This section expands on the key concepts referred to in the Principles.

2.1 Justifiability Principles (P1 and P2)

Justifiability

P1 Individuals or groups of individuals are not systematically disadvantaged through AIDA driven decisions, unless these decisions can be justified.

P2 Use of personal attributes as input factors for AIDA driven decisions is justified.

2.1.1 Group and Individual Fairness (P1)

Fairness can be broadly categorised into:

- **Group fairness.** Population groups are defined, and the outcomes between groups compared.
- **Individual fairness.** The notion of similarity in outcomes for similar individuals is defined, and outcomes between similar pairs of individuals compared.

Assessing fairness at too granular a level (e.g., for small groups of individuals, or at the multi features/ micro segments level) may be too complex to execute and can be limited by the collected features, privacy aspects or consent provided by individuals. As a result, assessing fairness at group level is a more common practice. We have included assessing both group fairness and individual fairness in the use case document 'Fairness in Predictive Underwriting'.

When assessing fairness at group level, FSIs need to be aware that compound effects may exist at the intersection of the attributes of certain groups (e.g., gender, ethnicity, disabilities, etc.), and a fairness assessment may need to be performed on selected combinations (say females who belong to a minority ethnicity), to enable the financial inclusion of such groups which may be at greatest risk. An assessment of every possible combination of groups may not be pragmatic, so a prioritised selection of at risk groups is recommended in consultation with stakeholders.

Fairness can be assessed quantitatively using a **Fairness Metric** - a mathematical definition of "fairness" that is measurable. Examples include False Negative Rate Ratio and Disparate Impact. Guidance on commonly used fairness metrics to quantify group and individual fairness is provided in appendix section 8.1 and 8.2. In many cases, different fairness metrics cannot be satisfied simultaneously and selecting the optimal metric for a use case is a non-trivial task. Different approaches exist in the literature for identifying the best metric for a given use case. An example is shown below from Aequitas (Figure 2.1), an open source bias audit toolkit. An application of this approach refined for the insurance sector is in the Phase 2 Illustrative Case Studies Document.

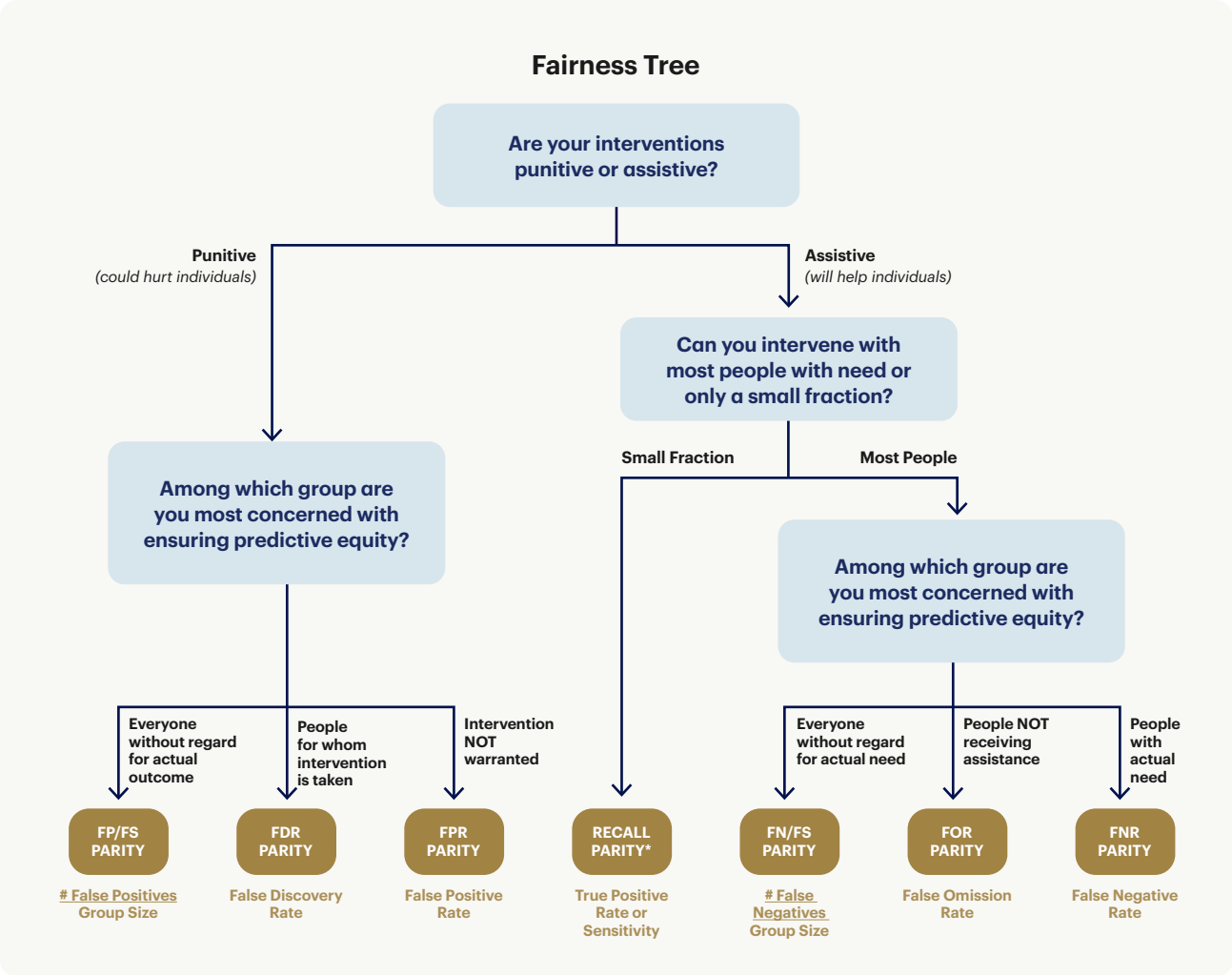


Figure 2.1 An example approach for selecting the most appropriate fairness metric, taken from Aequitas – an open-source bias audit toolkit.

Another example framework is the Fairness Compass which can be found in 'Towards the Right Kind of Fairness in AI'. Some fairness concepts specific to insurance are provided in appendix section 8.11.

2.1.2 Systematic Disadvantage (P1)

Systematic disadvantage refers to a negative impact from unevenly distributed harms or benefits resulting from AIDA driven decisions across individuals or groups of customers. If the use of an AIDA system for decision making leads to a particular group experiencing more harms or benefits than other groups, this could be deemed unfair unless the outcome can be justified. FSIs should decide, with sound justification, how to measure such potential systematic disadvantage (often using the relevant fairness metric), and to what extent the distribution of harms and benefits should be adjusted, if at all, to meet desired fairness objectives.

Additionally, the impact of the harms and benefits of a system from the perspective of affected individuals could also vary considerably from that of the FSI. For example, incorrectly auto declining a homeowner insurance application can cause substantial harm to the customer, who will be unable to protect their home. However, the harm for the insurer is only the loss of income from an individual policy, which will likely be immaterial relative to the overall business portfolio. A similar imbalance of harms would be experienced if a bank incorrectly declined a mortgage application on the basis of an automated assessment. It is therefore vital for FSIs to consider harms and benefits through the lens of their customers, as well as through their business lens, to understand the true impact of erroneous outcomes.

It is important to distinguish between

- **Systematic disadvantage** stemming from the operations of AIDA systems; and
- **Systemic disadvantage** stemming from macro level imbalances that could be reinforced by the usage of an AIDA system

One example of the latter is the hiring practices around the world, which often led to women being paid less than men for equal work. These practices could be reinforced by a poorly designed AIDA system.[15]

The design of AIDA systems should be undertaken with the aim of minimising unjustified systematic disadvantage, while also including considerations to avoid reinforcement of systemic disadvantage. Avoiding reinforcement of systemic disadvantage should be considered when determining at risk groups.

2.1.3 Justifications for System Decisions (P1)

For the cases where systematically disadvantageous decisions emanate from AIDA systems, FSIs should justify these decisions by demonstrating that they are reasonable. Otherwise, they should make changes, consistent with the Principles and aligned with internal organisational values, laws and regulatory requirements. Furthermore, in line with the Transparency FEAT Principles, FSIs are encouraged to explain their decisions to affected individuals, as part of their duty towards fair and transparent processes and outcomes (please refer to the Transparency Principles P13 and P14 in Transparency Methodology document).

2.1.4 Personal Attributes (P2)

Personal attributes are features about individuals that should not be used as the basis for decisions without reasonable justification. Personal attributes can be deemed high risk in fairness considerations, especially when a group of people characterised by those attributes is being considered. The use of personal attributes need to be justified and should not be used if there is no proven impact on improving fairness or creating benefits. Personal attributes and personal data are not the same.

- **Personal attributes** are defined by FSIs in the context of each specific use case and at a minimum covers any personal data that may be included in the AIDA system, as defined in relevant data protection and anti-discrimination laws, and can also include non-personal data.
- **Personal data** is defined as data, whether true or not, about an individual who can be identified either from that data or from that data and other information to which the organisation has or is likely to have, access.

Data attributes should be collected and processed in a lawful and fair manner.

Therefore, if the AIDA system includes personal or sensitive personal data, this data should be limited to what is reasonably necessary to meet the legitimate purpose.

In identifying personal attributes, FSIs may be guided by the following considerations:

- **Law, case law or regulations.** Relevant data governance or anti-discrimination laws, at local, regional, and global levels. This includes general and sector-specific regulations. Singapore examples include:
  - Constitution of Singapore, Part IV on Fundamental Liberties.[16]
  - Personal Data Protection Act, 2012, Singapore.[17]
  - Singapore Personal Data Protection Commission's Guide to DPIAs 2017.[18]
  - MOH-LIA Moratorium on Genetic Testing and Insurance, 2021, Singapore.
  - United Nations Convention on the Rights of Persons with Disabilities (UN CRPD)[19] which Singapore ratified in 2013[20], with an exception that was subsequently withdrawn in 2021.[21]
- **Emerging AIDA policies and frameworks.** These considerations are constantly evolving and would need to be re-evaluated by FSIs based on new developments (please see the appendix 8.12).
- **Socio-cultural acceptance.** This could be based on a range of methods, such as customer consultation, focus groups, expert advice, and a review of the appropriate literature. Particular attention should be paid to the sentiments of minority groups, as these might differ from those of the majority.
- **Market practice.** Reference can be made to existing practices by FSIs in the sector and any ongoing discussions on related topics.
- **Organisational values.** Where the FSI has multinational operations, due consideration should be given to both global principles and local contexts as well as an organisation's internal guidelines (e.g., information classification).

Once identified, personal attributes should be classified so that the appropriate action can be taken in the context of the AIDA system. A proposal for achieving this is outlined below, and in figure 2.2.



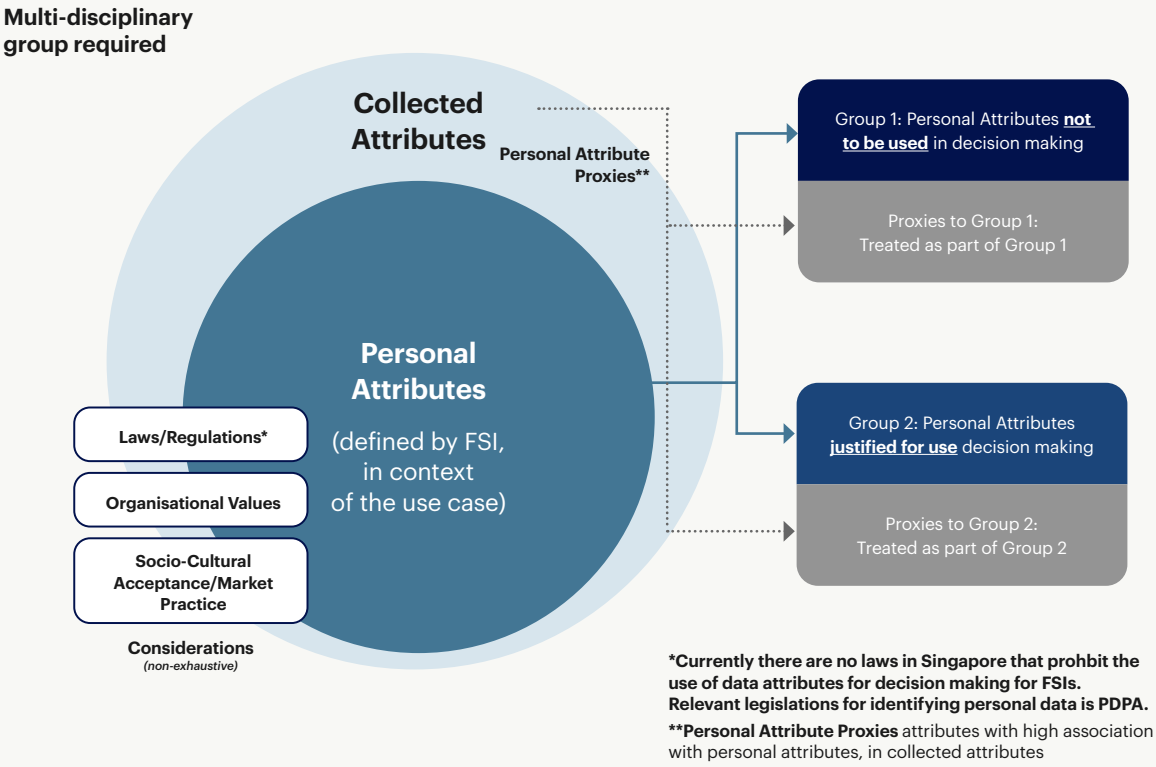


Figure 2.2 Identifying Personal Attributes

The personal attributes can be classified into two buckets, with associated recommended actions as follows:

- **Group 1: Personal attributes not to be used in decision making.** These attributes would be excluded from decision making and monitored for fairness across groups segregated by these attributes. At the minimum, this would include attributes mandated by law or regulation to be excluded for the use case under consideration.
- **Group 2: Personal attributes justified for use in decision making.** With a clear justification for inclusion documented, these attributes may be included in the decision process of the AIDA system. FSIs may decide to also monitor these attributes for fairness, depending on the specific use case context.

The personal attributes which are justified for use in AIDA decision making are specific to the use case in question. For example:

- Financial transaction data such as debt/income ratio and days past due on existing debt are accepted to be a determinant of credit risk, making it acceptable for use in AIDA systems that assess credit risk.
- Declaration of medical history, despite it being highly personal, is accepted as part of the underwriting process for life insurance given its strong links to future mortality risk.

Non-personal attributes could potentially act as proxies for personal attributes where statistical dependencies exist (meeting an association criterion as defined by the FSIs). The use of such data in the AIDA system's decision process could therefore pose a risk of indirect discrimination with respect to the related personal attributes.

It should be highlighted that proxies identified to Group 1, including Group 2 Personal Attributes, should also be treated as personal attributes and not be used for decision making. Proxies identified to Group 2 (other than existing Group 1 personal attributes) may be justified for use in decision making.

2.1.5 Justification for Use of Personal Attributes (P2)

Justification for use of personal attributes in decision making could include:

- **Need to comply with legal or regulatory requirements for the AIDA enabled product or service.** For example, if a product may only be offered to citizens of Singapore, then citizenship information may be used according to local regulations.
- **Essential to the basic proposition of the product or service.** For example, the year of birth is needed when providing child protection insurance.
- **Material for fairness objectives.** Where including the attribute allows FSIs to explicitly tune the models or calibrate outputs to enable a more equal distribution of benefits/ harms.
- **Knowledge of historical biases or prejudices.** Where there is reason to suspect that historical data used to train the models is biased from prior data or decision systems, operational use allows FSIs to control for fairness metrics, avoiding unintended reinforcement of such bias.
- **Modifiable nature of attributes.** Where modifiable attributes within the control of individuals have material influence on the actual outcome.
- **Material for commercial objectives.** Where including the attribute allows FSIs to conduct substantially more accurate predictions, such that the benefits of inclusion outweigh the drawbacks. The impact on key fairness metrics of inclusion on commercial grounds, along with justification for the trade-offs, should be documented.

It is important to note that because of the dynamic nature of data driven approaches and the use of AIDA systems, there remains a continued discussion on the definition and use of certain personal attributes - with differences across regions and FSI needs to monitor latest regulatory developments in this field.

2.2 Accuracy and Bias Principles (P3 and P4)

Accuracy and Bias

P3 Data and models used for AIDA driven decisions are regularly reviewed and validated for accuracy and relevance, and to minimise unintentional bias.

P4 AIDA driven decisions are regularly reviewed so that models behave as deisgned and intended.

2.2.1 Regular Review (P3 and P4)

The appropriate frequency for regular review post deployment is context specific, and should be commensurate with the AIDA system's risk level and with relevant laws and regulations. In addition to specific timeframes during which the review needs to be completed, FSIs should set thresholds for critical metrics, such as accuracy and error rates that, if breached, would trigger off cycle reviews. Although relevant data checks are carried out during the development phase, there should be ongoing, regular data checks following system deployment, such as model input distribution (numerical features) or frequencies (categorical features), as well as missing value checks, to monitor for data drift. It is also important that the review process is robust throughout the AIDA system development cycle.

2.2.2 Accuracy (P3)

Accuracy refers to how well a model's predictions match the true state of the world. Under the Principles, this term need not necessarily refer to the conventional data science definition of accuracy (for instance, accuracy of classification models calculated as the ratio of correctly predicted observations to the total observations). Rather, FSIs should decide on the most appropriate measure for accuracy based on the context and purpose of the AIDA system.

2.2.3 Relevance (P3)

Relevance refers to validation that the data used constitutes a material contribution towards achieving the AIDA system's legitimate commercial and fairness objectives. Relevance can stem from:

- Data fields being direct components of the AIDA system's predicted outcome.
- The predicted outcome being statistically or causally associated with the data fields.
- The data fields providing important information about the population groups to which individuals belong.

2.2.4 Unintentional Bias (P3)

Unintentional bias refers to systematic disadvantage that the FSI is not aware of, resulting from undetected bias in the data or model algorithms or how they are used. Sources of bias include data, model and deployment decisions made by humans that reflect cognitive biases such as implicit biases against particular groups, confirmation biases and decision fatigue. These can manifest in quantitatively measurable biases that can be observed during AIDA system development.

Bias can enter the AIDA system at different points in the development lifecycle as shown in Figure 2.3 (non-exhaustive), where common biases are mapped against the AIDA System lifecycle.

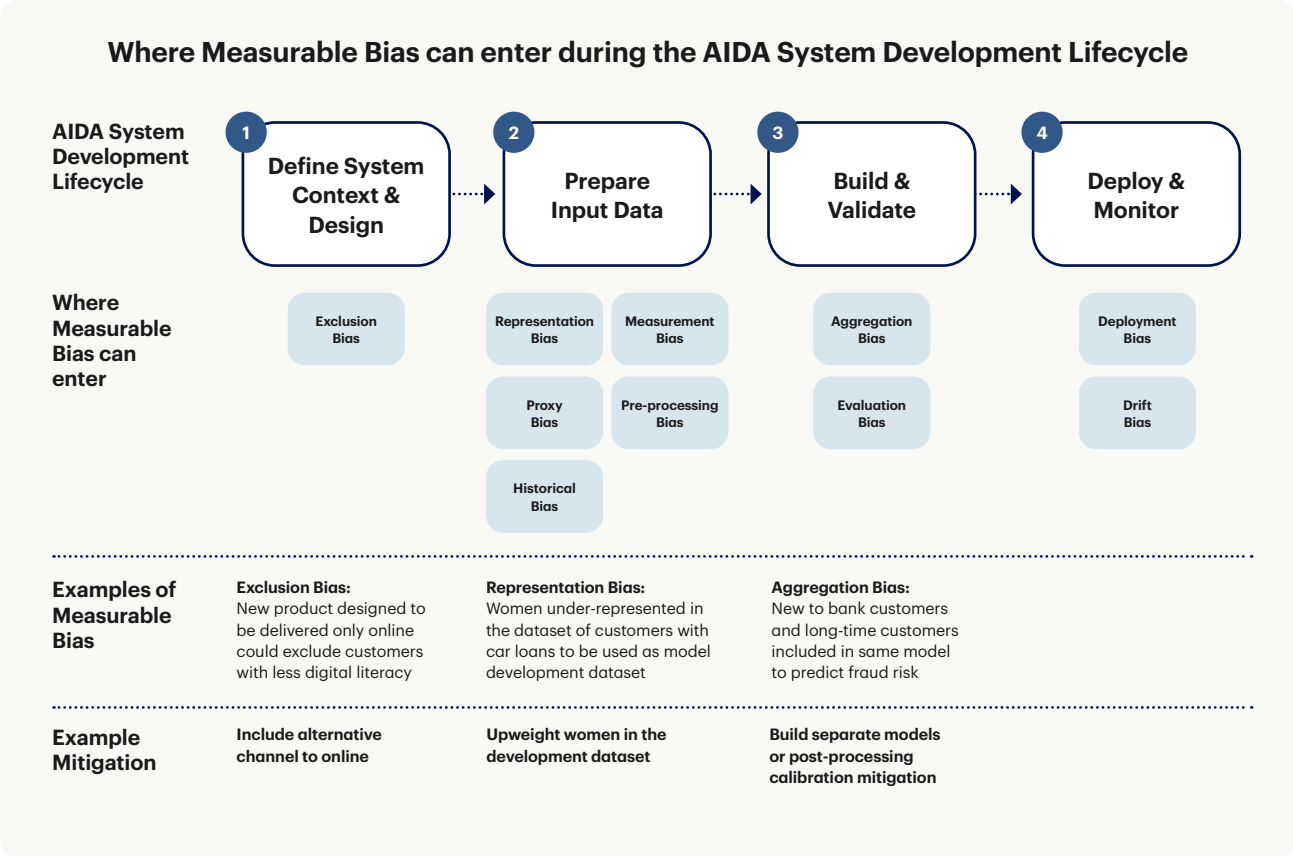


Figure 2.3 Common biases appearing in an AIDA model development lifecycle

At Step 1: Define System Context & Design, a type of measureable bias that can enter the system is:

- **Exclusion Bias.** Arises when certain groups are excluded from the development dataset.

At Step 2: Prepare Input Data, common types of measureable data bias that enter are:

- **Representation bias.** Occurs when certain groups are underrepresented in a data set which causes the effectiveness of model training to be hampered. Under-representation can be caused by selection bias during data collection, or simply by the fact that the affected group is a small portion of the overall population.

- **Measurement bias.** Arises when there is systematic or non-random error in the collection of data, and can occur on input variables and target labels on which the AIDA system operates. Measurement bias can originate from misreporting or nondisclosure by customers or distributors, creating erroneous data and information asymmetry.
- **Proxy bias.** Occurs in cases where input or target variable is estimated using a proxy variable which is systematically different from the variable of interest.
- **Pre-processing bias.** Arises during data pre-processing in model development, when an operation (e.g., missing value treatment, data cleansing, outlier treatment, encoding, scaling or data transformations for unstructured data, etc.) causes or contributes to systematic disadvantage. Examples include:
  - **Missing data.** Depending on whether data is Missing Completely at Random (MCAR), Missing at Random (MAR) or Missing Not at Random (MNAR), appropriate techniques should be used to prevent the introduction of bias through data imputation.
  - **Labelling.** A human assisted risk labelling process introduces subjectivity and increases the risk of cognitive bias from human led decisions.
- **Historical bias/negative legacy bias.**[20] Occurs when bias, human or otherwise, present in prior decisions is reflected in the training data. Without adjustments, these biases will be perpetuated in new decisions from the AIDA systems.

At Step 3: Build and Validate, types of measureable bias include:

- **Aggregation bias.** Occurs when a one size fits all model is employed for groups with different risk distributions.
- **Evaluation bias.** Arises when the testing or external benchmark populations do not equally represent the various parts of the user population.

And at Step 4: Deploy and Monitor, examples of measureable biases that can enter are:

- **Deployment bias.** Occurs when there is a mismatch between the problem a model is intended to solve and the way in which it is actually used.
- **Drift bias.** A type of bias that can occur over time post implementation. This can be due to:
  - **Data drift.** The circumstances of individual or general population characteristics change over time, whereas the system's training data remains static.
  - **Concept drift.** The statistical properties of the target variable to be predicted change over time due to factors including population risk trends, technological advancements, and evolving behaviour patterns. As a result, relationships between factors and outcomes shift. For example, a risk assessment of an uninsured population can be different to that of an insured population, because the latter may change their behaviour in light of the fact they are insured.

2.2.5 Minimise Unintentional Bias (P3)

The Principles require FSIs to **minimise unintentional bias** by incorporating reasonable steps in the AIDA system development process to identify and address potential sources of bias throughout AIDA system development and deployment, so as to avoid inadvertent systematic disadvantage. While there is no one size fits all approach and no single change that can avoid unintentional bias, steps can be introduced at each stage of AIDA development to identify and mitigate both data and human biases.

Qualitative Steps to minimise unintentional bias by preventing cognitive biases from occurring can include:

- **Process enhancements.** Introducing behaviourally informed interventions that avoid biases being introduced at source (e.g., data collection enhancements, checklists and appropriate challenge and review techniques) throughout the AIDA development lifecycle
- **Environmental Change.** Ensuring the working environment, culture and incentives do not drive behaviours that do not align with fairness best practice (e.g., reducing decision fatigue by ensuring appropriate working practices and timelines during development)
- **Diverse representation.** Proactively including a variety of thought and experience during development e.g., through diverse development teams and where appropriate end user feedback

Quantitative steps to minimize unintentional bias already in the data or model include technical bias mitigation methods. These methods can be implemented during the AIDA system development with outcomes quantitatively measured for effectiveness against the fairness metrics. The methods can be applied at the preprocessing, in-processing and post processing stage. More information on these methods can be found in Appendix 8.5.

2.2.6 Alignment of Decisions with Model Design and Intent (P4)

FSIs should continually monitor AIDA driven decisions against their models' design and intent, from both commercial and fairness perspectives. Specific to fairness, FSIs should regularly check that their selected fairness metrics remain relevant to the context and purpose of their AIDA systems, and that their defined fairness objectives continue to be met.

03 FEAT Fairness Principles Assessment Methodology ("Fairness Methodology")

Understanding an AIDA system's alignment with the Principles is an ongoing process due to changing data inputs, usage context, customer behaviour, business priorities and other factors. This section presents a methodology for fairness assessments that aligns with the Principles, using a series of guiding questions to be answered by FSIs and presented to AIDA system assessors for fairness evaluation.

The application of the Fairness Methodology is illustrated through case studies on customer marketing and credit scoring in Veritas Phase 1,[23] and in Phase 2 through the use case of predictive underwriting for cross-selling life insurance.

In the following subsections we show how the Fairness Methodology can be embedded into a typical AIDA system development lifecycle, before introducing the five parts of the Fairness Methodology, the questions in each part, and considerations to assist in developing responses to the assessment questions. Please note that the examples provided in this section are intended only as an aid to comprehension and do not provide the full context of the relevant use case. For a more detailed application of the Fairness Methodology to use cases, please refer to the Credit Score Fairness Assessment Case Study document from Phase 1 (for Banking) and the Predictive Underwriting Fairness Assessment case study document from Phase 2 (for Insurance). Appendix 1 contains a technical reference for further detailed discussions on these considerations.[24] Where appropriate, FSIs should include additional questions or considerations specific to the context and objectives of the AIDA system under consideration.

FSIs should adhere to all legal and regulatory requirements when undertaking activities relating to the collection, use, processing or disclosure of personal attributes involving personal data by conducting a Data Protection Impact Assessment (DPIA) before the FEAT Assessment takes place. The practices suggested in this Fairness Methodology are for general information and not exhaustive.

The depth of the fairness assessment should be customised to be commensurate with the risk level of the AIDA system in question and balanced against the additional costs that eventually get passed on to end customers. Customisation can include the questions and considerations answered, the extent of analysis and level of documentation – see section 4 for more detail.

3.1 Embedding the Fairness Methodology into the AIDA System Development Lifecycle

The diagram below outlines the fairness questions and how they can be embedded into the AIDA system lifecycle. The 18 questions from Veritas Phase 1 have been mapped to four generic FEAT checklist questions (relevant to all of FEAT) and 12 FEAT Fairness-specific checklist questions – see appendix 8.13 for the detailed mapping. These questions enable the FSI to translate their organisational values on fair and responsible AIDA driven decision making into practical steps through quantifiable metrics and thresholds. Embedding into the AIDA system lifecycle enables FSIs to proactively develop fair AIDA systems by design.

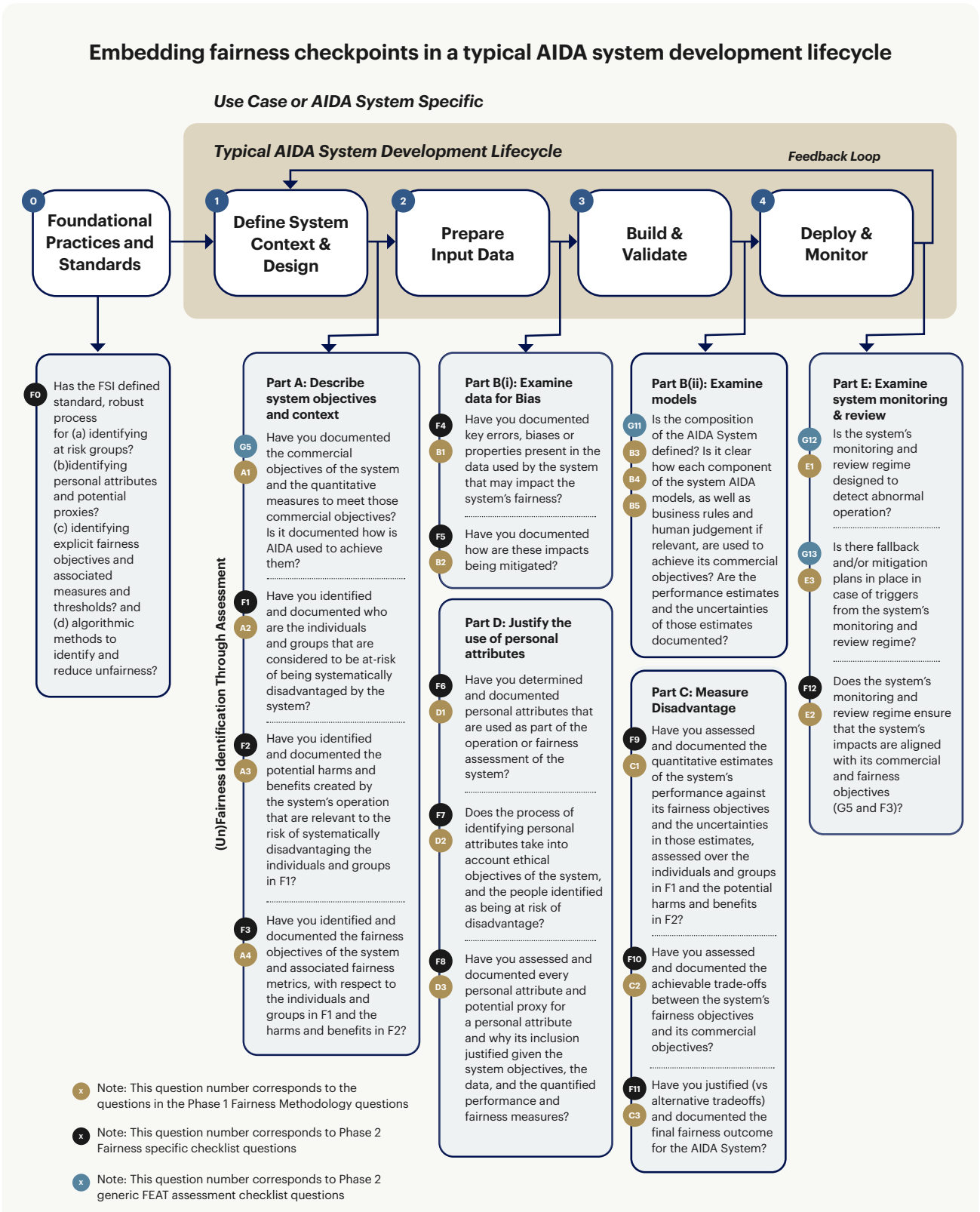


Figure 3.1 Mapping the Fairness Methodology to a typical AIDA system development lifecycle (Adapted based on Veritas Phase I published by MAS)

3.2 Step 0: Foundational Practices and Standards (Fairness Specific)

As the definition, quantification and target levels of fairness can vary by use case, the absence of guidance on this will result in high levels of subjectivity, as each AIDA project team could potentially use a different decision framework.

Therefore, in addition to embedding the Fairness Methodology in the AIDA Development lifecycle, it is recommended that FSIs have an **internal foundational set of practices and standards** approved via the relevant governance structures in line with regulatory requirements. Defining a set of internal, use case agnostic fairness standards upfront can provide a level of consistency across all use cases in the FSI while still allowing room for tailoring based on materiality. Such standards should be guided by prevailing legal and regulatory requirements, the FSI's own data/ AIDA ethics principles, and learnings from prior work on fairness in the organisation (if any) and should include the following:

1. **Guidance to customise the depth of the fairness assessment.** To be commensurate with the fairness risk level of the AIDA system under consideration. Such customisation could relate to the assessment process itself, the level of detail of the assessment (including which methodology questions and/or considerations are answered), or the parties involved.
2. **Internal framework for identifying at risk groups, personal attributes and their proxies.** With guidelines on associated actions in terms of exclusion or inclusion with justifications, as well as what an acceptable justification looks like.
3. **Standard process to explicitly define fairness objectives.** By type of use case and inventory of associated appropriate measurable fairness metrics.
4. **Guidance for determining acceptable deviation thresholds.** For measures of fairness and associated **trade-offs** with system performance or other fairness measures.
5. **Reference toolkit(s).** To conduct unfairness identification checks and algorithmic mitigation methods for use across the stages of system development.

Bullet 1 above is covered in more detail in section 4.2, and bullets 2-5 are covered in Fairness assessment question F0 below. The guidelines should be framed once when setting up the standards, and then reviewed periodically to enable ongoing alignment with evolving technology (e.g., alternate fairness measures or algorithmic mitigation methods), regulatory thinking (e.g., new guidance on anti-discrimination or personal information protection) and internal or customer feedback (e.g., disproportionate overrides by users, redressal requests or complaints across target customer population groups). In addition, academic conferences on AIDA that include fairness tracks and academic literature on evolving best practices for evaluating AIDA fairness can help with improved assessment approaches. One approach to F0 would be to define this for the key AIDA systems in use by the FSI initially and then in periodic updates expand the scope for additional relevant AIDA systems.

F0 **Has the FSI defined standard, robust process for (a) identifying at risk groups? (b) identifying personal attributes and potential proxies? (c) identifying explicit fairness objectives and associated measures and thresholds? and (d) algorithmic methods to identify and reduce unfairness?**

3.3 Step 1: Define System Context & Design

3.3.1 Part A Describe System Objectives & Context

G5 **Have you documented the commercial objectives of the system and the quantitative measures to meet those commercial objectives? Is it documented how is AIDA used to achieve them?**

[This question refers to question A1 in the Phase 1 Fairness Methodology questions]

Explicitly stating the commercial objectives of the AIDA system, along with clear documentation on its structure and design, serves as the foundation for responses to the subsequent questions within the Fairness Methodology. This process enables deployment of the system to be contained within the specific context for which it was designed, thereby maintaining validity of the fairness assessment.

No.	Key considerations	Proposed general guidance *(examples italicised)*
1	What are the primary and secondary commercial objectives of the system?	Commercial objectives refer to the outcomes to improve the business's bottom line that are intended to be driven by the given use case. They can be expressed in terms of items like sales, revenue, profit, ROI, turnaround time, customer satisfaction/NPS, straight-through processing (STP) rates, lower acquisition costs, etc. These objectives may be classified as Primary or Secondary (ancillary) goals based on the use case context. The primary commercial objective is used when determining trade-offs with the fairness metric, where required.
2	How is AIDA used to achieve these commercial objectives?	Establish a clear link to demonstrate how AIDA system outputs will help improve the stated commercial objectives. *For instance, to reduce the turnaround time for granting access to a financial product, the AIDA system accelerates the identification of eligible individuals through data driven techniques that are applied to a wide range of attributes the FSI has collected with customer consent.*
3	Are there internal or external constraints that affect the ability of the system to achieve these objectives?	An example of an external constraint could be missing information when customers do not disclose similar services/products with other FSIs (e.g., non-disclosure of a loan with another bank, or a condition managed or disclosed with another insurer). Another example of an external constraint would be that FSIs must comply with antitrust and competition laws (see section 1.2.3)
4	What are the potential benefits (and downsides, if any) of the system for the FSI?	Indicate benefits of using the AIDA system in place of the non-AIDA or historic approach used by the FSI. Also note downsides (if any) of erroneous outcomes from such a system. *For instance, the benefits of a banking or insurance system could include data driven approaches for better inclusion, simplified customer experience, better and quicker services with automation, etc.* *Conversely, downsides could include financial losses due to the onboarding of high risk customers if the actual level of error in the AIDA system is higher than expected*
5	Are the structure and design of the AIDA system clearly documented?	Functional and technical details for the AIDA system should be documented (also including FEAT adoption aspects across the development lifecycle). Even at a high level, this can provide important actionable insights to minimise, mitigate or prevent potential causes of unfair operation. Examples include human design choices encoding biases into the AIDA system, or deploying an AIDA system outside the context for which it was designed and thereby creating unintended harms.

F1 **Have you identified and documented who are the individuals and groups that are considered to be at risk of being systematically disadvantaged by the system?**

[This question refers to question A2 in the Phase 1 Fairness Methodology questions]

Identification of the individuals and groups that will be affected by the AIDA system's decisions is the first step to analysing the potential harms and benefits they could experience. For identification of at risk individuals and groups, it is important to refer to laws, rules, regulations and guidelines in the jurisdiction or for the same industry in other regions, in addition to being informed by societal, cultural, or historic prejudices/biases in similar decision making.

A multidisciplinary team including the AIDA system owner, data scientists, business experts, legal, compliance and risk experts, data ethics and responsible AI experts, will enable a holistic assessment and minimise the risk of affected parties not being identified. This can be supplemented by reference to existing AIDA systems in use, and to relevant AIDA driven financial services in other countries.

Legal, regulatory, societal and organisational key considerations:

No.	Key considerations	Proposed general guidance *(examples italicised)*
1	Are there individuals or groups that are already disadvantaged, whose interaction with the system could perpetuate or reinforce that disadvantage (i.e., systemic risk)?	For the identification of at risk individuals/groups, it is important to refer to anti-discriminatory guidelines or protected classes, in addition to being informed by societal, cultural, or historic prejudices/biases in similar decision making.
2	Do the potential harms caused by the system result in greater impact for some individuals or groups?	It is important for FSIs to consider the impact on individuals or groups due to adverse decision by the system and particularly if the adverse decisions are erroneous. *For instance, the termination of a service - or revision of the service terms - due to a wrong AIDA model decision could have a bigger impact on the less resilient/lower income group of customers.*
3	For individuals, are there combinations of group memberships that may cause compound disadvantage?	Very often the features to be tested for fairness are not mutually exclusive - gender, marital status, ethnicity among others. Therefore, compound disadvantage may exist, and the fairness assessment may need to be performed on selected combinations (say females who belong to a minority ethnicity), to ensure financial inclusion of such sub-groups which may be at risk of disadvantage. Assessment of all combinations of groups may not be pragmatic or possible, so a prioritised selection of at risk groups by multi-disciplinary experts is recommended.
4	Are there individuals or groups identified as at risk in other countries who might be relevant for consideration in this context?	It is good practice to be aware of key individuals or groups that are considered at risk in other jurisdictions for similar use cases, to evaluate whether they also might be relevant to assess.

System considerations:

No.	Key considerations	Proposed general guidance *(examples italicised)*
1	Does the product or service offered through the AIDA system have eligibility criteria, restrictions or design factors that cause it to exclude use by some individuals or groups?	*For instance, automation of a distribution channel or digitisation of services could exclude customers with less digital literacy or with poorer online, digital data access.*
2	Does post-deployment data from AIDA systems currently in production indicate potential individuals and groups who may be subject to greater risk of harm?	Comparing stratified outcomes by population subgroups to analyse variance from the current live environment data may uncover current operational processes that could subject specific groups to greater risk of harm *(e.g., the impact of human manual adjustments if more prevalent for specific groups)*.

F2 **Have you identified and documented the potential harms and benefits created by the system's operation that are relevant to the risk of systematically disadvantaging the individuals and groups in F1?**

[This question refers to question A3 in the Phase 1 Fairness Methodology questions]

To arrive at general guiding principles to detect specific harms and benefits that individuals and groups identified to be at risk (F1) could be subjected to is to combine responses and assessments of:

- Commercial objectives of the system (G5).
- Individuals and groups that are considered to be at risk of being systematically disadvantaged by the system (F1).
- Layering in customer views (where feasible).

The impact of these harms and benefits will in turn inform decisions relating to fairness objectives.

No.	Key considerations	Proposed general guidance *(examples italicised)*
1	What are the customers' views about the benefits or risks of the system?	One way to gain an understanding of customers' views is to run focus groups or customer surveys on the proposed offering that could include their views on cost/benefits. The FSI can highlight if the views differ between groups identified as at risk of disadvantage in F1 and the rest of the population, and use this information as an input when identifying the benefits and harms of the AIDA application.
2	What steps were taken to ensure important harms and benefits do not go undetected? Are there potential harms that may have been missed by customer consultation?	It is recommended to have a multidisciplinary team, comprising the AIDA system owner, data scientists, business experts, legal, compliance and risk experts, data ethics and responsible AI experts, to provide input on potential harms. This can supplement customers' views.

No.	Key considerations	Proposed general guidance *(examples italicised)*
3	Which are the most important harms to consider for the groups identified as at risk of disadvantage in F1? Why?	If the AIDA system outcome is assistive (such as assessing if an individual should get automated access to a financial product) then the most important harms to consider usually are whether at risk groups are incorrectly denied assistance more often than others. Whereas if the AIDA system outcome is punitive (such as assessing if an individual is fraudulent), then most important harms to consider usually are whether at risk groups incorrectly receive the punitive outcome more often than others.
4	Which are the most important benefits to consider for the groups identified as at risk of disadvantage in F1? Why?	Checking that the distribution of benefits to at risk groups is comparable to the target population is more relevant for assistive AIDA systems than for punitive ones. Additionally, there may be wider gap in circumstances of at risk groups from the preferred state of being, making it even more important for the benefits of the financial products/services to reach these groups.
5	Do the potential harms or benefits impact the groups identified as at risk of disadvantage in F1 in other indirect, longer-term ways?	*For example, banks and insurers should assess the ability of groups to afford the payment of services over the long term, once selected by the AIDA system.*
6	What are the opportunity costs to the groups identified as at risk of disadvantage in F1 associated with particular outcomes?	*An example of this would be the opportunity cost of allocating the financial resources to finance the banking or insurance services against other services that may be particularly important for the long-term resilience of groups identified as at risk in F1, (e.g., investment or savings).*



F3 Have you identified and documented the fairness objectives of the system and associated fairness metrics, with respect to the individuals and groups in F1 and the harms and benefits in F2?

[This question refers to question A4 in the Phase 1 Fairness Methodology questions]

With an understanding of the individuals and groups affected by the system, and the harms and benefits they might experience, FSIs can determine fairness objectives which are appropriate to the context and aligned with internal organisational values.

No.	Key considerations	Proposed general guidance (examples italicised)
1	What are the organisational principles, values, standards (local and global) or guidelines relevant to fairness considerations?	Refer to the company vision/ mission/ purpose and internal guidelines. Multinational FSIs should consider both global and local standards. A process to start from organisation specific values to principles, including fairness, to measurable considerations is included in the Ethics and Accountability Methodology document.
2	What are the identified fairness objectives for the system?	The fairness objective of AIDA system is use case specific and should be defined by the AIDA system Owner (usually the FSI process business owner). One way fairness objectives can be defined is using a fairness decision tree, focused on specific at risk groups, where each decision point question answered helps to narrow down to the relevant fairness objective, and relevant fairness metric.
3	Are fairness objectives consistent across similar use cases/decision types (or if not, are the reasons understood and documented)?	A standard process to determine fairness objectives for each AIDA system across an organisation enables consistency. Comparison of chosen fairness objectives against other similar AIDA systems already deployed can also help to achieve consistency.
4	Are the fairness objectives appropriate for the commercial objectives and decision outcomes of the system?	The fairness objectives of AIDA system are use case specific and can be different for the same AIDA system when used for different purposes. For example, if the same AIDA system used to fast track application acceptance is then used for application rejection, this will change the choice of fairness objective.
5	What is the process by which the fairness measure(s) (or metric(s) are selected for the system?	Adopting a standard, robust assessment process to identify the key important fairness metrics for the use case and decision type the AIDA model is being developed for helps to ensure the most relevant fairness metrics are chosen consistently by the AIDA system owner/developer. An example of a "fairness metrics decision tree" that can be used for this process is illustrated in the use case document. In addition, calculating a consistent set of standard fairness measures to check for any significant imbalances can mitigate the risk that a harm or benefit was missed in an initial analysis. The choice of fairness metrics and thresholds aligned with system's commercial objectives should be done through upfront selection of metrics (before the model is developed) instead of selection post assessment, when there is a danger that the choice of measures ex post is selected based on showing the least unfairness.

6	What are the reasons for the choice of metrics, with respect to the overall commercial and fairness objectives?	Ensure that the fairness metrics chosen align with the overall commercial and fairness objectives by having a clear understanding of what "fairness" means with respect to the use case. Mathematical formulation of a fairness metric is guided by the choice of fairness objectives in A4. (Refer to appendix 8.1 for commonly used fairness metrics to measure identified fairness objectives)

3.4 Step 2: Prepare Input Data

3.4.1 Part B(i): Examine Data for Bias

F4 **Have you documented key errors, biases or properties present in the data used by the system that may impact the system's fairness?**

[This question refers to question B1 in the Phase 1 Fairness Methodology questions]

Data is an input to an AIDA model, and the context on how the data was collected and prepared can have a significant impact on the level of bias in the data. This could cause systems to learn and replicate or amplify such biases within the system, resulting in unfair outcomes. FSIs should have a strategy or set of standard checks to avoid creating or reinforcing unfair bias in the AIDA systems due to the use of input data. It is assumed FSIs are following standard data minimisation and proportionality principles when collecting and processing the data to be used in AIDA systems in order to protect the privacy of individuals.

No.	Key considerations	Proposed general guidance *(examples italicised)*
1	For each group at risk of being systematically disadvantaged (F1), what data are used to determine membership to those groups?	Check availability, permissibility and justifiability of data (features) to identify groups at potential risk of disadvantage. Validate credibility and the context of the source(s) of information used to identify membership to these groups.
2	How is privacy, confidentiality and compliance ensured while using personal data to determine membership to groups identified as at risk of disadvantage in F1?	As FSIs look to leverage and share their sensitive data assets for maximum economic return, ensuring privacy, confidentiality and compliance is a must. Emerging Privacy Enabling Technologies that can be used include differential privacy, de-identification, data anonymisation and data masking, homomorphic encryption, federated learning, secure multiparty computation synthetic data, transfer learning and others. If personal data is used, the FSI should conduct a Data Protection Impact Assessment, and understand and manage the risk where individuals and groups might be deprived of exercising control over their personal data. Risks may vary in probability and severity. The assessment should provide the measures envisaged to address the risks, including safeguards, security measures and mechanisms to ensure the protection of personal data and to demonstrate compliance with applicable data protection laws and regulations.
3	Is there adequate representation of relevant groups in the target population group of the service and product, especially those identified as at risk of disadvantage in F1?	Due to data/population characteristics or sampling procedures, it may be observed that different groups may be under- or overrepresented in a dataset. For certain population subgroups (within the target group), inadequate representation in training data may cause higher proportion of erroneous outcomes, as model may not learn their characteristics adequately.

4	Are there potential sources of systematic measurement errors in the data?	In general, these errors are consistent and are predictable with regard to the true value, due to inherent systematic inaccuracies in measurement of an attribute.
5	Are there measurable proxies for unavailable or unmeasurable information being used in the data? If so, is the effectiveness of proxy estimation similar across different groups of the population?	The measurable proxies used in place of non-measurable properties are imperfect and may introduce fairness risks when they are less suitable for some groups compared to others.
6	Could data pre-processing techniques introduce additional bias into the input data?	Pre-processing of input data (such as missing value treatment, data cleansing, outlier treatment, encoding, scaling, and data transformation for unstructured data, etc.) may involve decisions that impact system outcomes. *For example, if an at risk group has more missing data than other groups, then imputation of that data may lead the system to disadvantage that group.*
7	Are there potential sources of historic decision bias reflected in the data?	Historic bias can be identified by analysing personal attributes in collected data, which may correspond to groups of individuals at risk of disadvantage for known historic prejudices. *For instance, women have historically received lower salaries than men in the same jobs and accounted for a greater share of part time jobs. This can adversely affect FSI's views on the future earnings trajectory of this group and impact the eligibility of women for certain financial products in both banking and insurance.*



F5

Have you documented how are these impacts being mitigated?

[This question refers to question B2 in the Phase 1 Fairness Methodology questions]

When identified, potential data biases should be considered and addressed where relevant prior to inputting the data into AIDA system decisions. This should minimise their effects and mitigate the resultant disadvantageous impacts to affected individuals and groups.

No.	Key considerations	Proposed general guidance *(examples italicised)*
1	How is representation bias mitigated?	Methods to improve representation include collecting more data, particularly that of underrepresented groups, and statistical techniques to balance representation, such as data reweighing, up-sampling or down-sampling. Refer appendix 8.5 for more information on mitigating representation bias
2	How is measurement bias mitigated?	Potential methods for reducing measurement bias are to: • Obtain another, more reliable source of data. • Improve quality of instruments/sources of data collection in light of related best-practices. • Train people who collect/support data collection or generate labels/ annotations. This can be done by 1) Introducing interventions at the moment of interaction between humans and AIDA systems (i.e., when creating a model, as opposed to documenting ex-post) and 2) creating decision hygiene checklists that create new ways of preventing AIDAs from being affected from human led biases.
3	How is proxy bias mitigated?	The most common approach to mitigate proxy bias is to change to a more relevant proxy, where the new proxy feature comes with an evidence based justification of causality, rather than potentially blunt substitution where features have a mere association with intended property to be measured.
4	How is pre-processing bias mitigated?	Bias due to pre-processing such as missing value imputations or outlier treatment or sampling, may be mitigated through first understanding the reason behind anomalies and fixing the root causes. Another potential way to mitigate is augmentation or guidance from external data to make informed data treatments.
5	How is historic bias mitigated?	Some methods to mitigate historic bias are gathering more appropriate (potentially recent) data and adjusting the selection criteria for the disadvantaged groups, building synthetic datasets that remove existing data bias, and testing AIDA model biases.

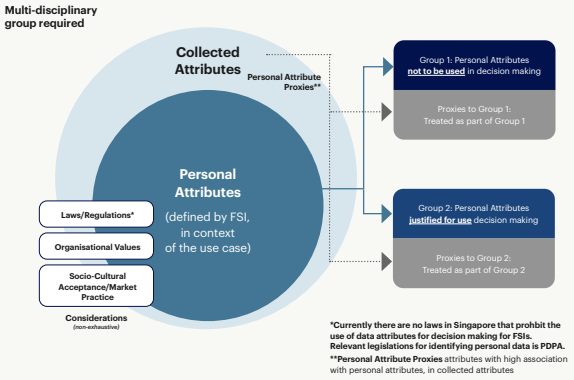
3.4.2 Part D: Justify the Use of Personal Attributes

F6 **Have you determined and documented personal attributes that are used as part of the operation or fairness assessment of the system?**

[This question refers to question D1 in the Phase 1 Fairness Methodology questions]

By first listing out the personal attributes used, either towards the AIDA system outcomes or in the fairness assessment, FSIs can systematically demonstrate the related considerations and justify their inclusion. Potential personal attributes that could be beneficial to include but are not present in the data are also considered to guide potential future action. (refer section 2.1.4 for discussion on identification of personal attributes)

	Key considerations	Proposed general guidance *(examples italicised)*
1	Which personal attributes are present in the data collected?	Following the list of considerations can help in schematic identification of personal attributes, as described in section 2.1.4.:
2	Which personal attributes are being used by the AIDA system for decision making (Group 2 in section 2.1.4)? Which of these are being monitored?	
3	Which personal attributes are excluded from decision making (Group 1 in section 2.1.4)? Which of these are being used for fairness assessment or monitored?	
4	Are there non-personal attributes in the data, which can act as proxy to "personal attributes excluded from decision-making"? Have those been addressed or excluded from decision making?	
5	Which additional personal attributes not available in the data are likely to be beneficial for use, in either decision making or fairness assessment?	
6	What options are there to collect and use these additional personal attributes within the constraints of legislation, regulation, societal acceptance, market practice and organisational values?	
7	Did you consider ways to develop the AIDA system or train the model without or with minimal use of potentially sensitive or personal data?	



F7

Does the process of identifying personal attributes take into account ethical objectives of the system, and the people identified as being at risk of disadvantage?

[This question refers to question D2 in the Phase 1 Fairness Methodology questions]

FSIs should have in place internal guidelines on the process of identifying personal attributes. The guidelines should be applicable across AIDA systems, but customisable to suit the specific use case context. Referencing these guidelines and demonstrating how they were customised will provide sound justification for the listed personal attributes and assurance that a comprehensive identification process was conducted. The "ethical objectives" referred to in the question relate to those beyond regulatory requirements, such as social, cultural, and organisational values.

No.	Key considerations	Proposed general guidance *(examples italicised)*
1	Is there a standard process for identification of personal/non-personal attributes (e.g., in accordance with all law, case law and regulations)?	Having a standard, robust process for identifying personal attributes is important to ensure organisation-wide consistency, considering the following in the process: • Local and global laws and regulations • Socio-cultural context • Market practice • Organisational values
2	Does the definition of personal attributes vary by use-cases and/ or industry context and/or region under consideration etc.?	The definition of personal attributes may change depending on the use case, as an attribute considered personal in one use case may not be considered so in other. Regulations, laws and societal opinions may change with the region under consideration and so does the definition of personal attributes. *For example, credit scoring in the US is governed by the Equal Credit Opportunity Act (ECOA), however in Singapore there has not been an equivalent piece of legislation.*
3	Who are the key stakeholders involved in determining personal attributes?	The process to determine personal attributes should involve multidisciplinary stakeholders, to ensure fitness of design from diverse perspectives. These should include AIDA system owner, data scientists, business experts, legal and risk compliance professionals and data ethics and Responsible AI experts.

F8

Have you assessed and documented for every personal attribute and potential proxy for a personal attribute, why is its inclusion justified given the system objectives, the data, and the quantified performance and fairness measures?

[This question refers to question D3 in the Phase 1 Fairness Methodology questions]

Decisions around which personal attributes to include in the AIDA driven decision process, and which to assess for fairness, should be justified with demonstration that the decisions are reasonable. Explicit attention should be paid to proxy attributes to provide assurance that unintended proxy discrimination effects are minimised.

No.	Key considerations	Proposed general guidance *(examples italicised)*
1	What are the justifications for inclusion, for the personal attributes being used in decision-making?	Questions to ask include: • Is the inclusion of the data required to meet a specified objective or basic operational aspects of the system? • How does the inclusion impact commercial and fairness objectives? • Why is the trade-off between them acceptable? • What specific facts about local context and history support the inclusion? *For example, most financial products are not allowed to be offered to minors. Therefore, information on age is required as part of business rules to determine the AIDA system's decision.* Inclusion of personal data, as defined by regulation, requires a higher standard of justification to demonstrate the legitimate interest in including it in the AIDA system compared to non-personal data.
2	What are the justifications for the personal attributes being used in decision making, but NOT being used for the fairness assessment/monitoring?	Clear rationale should be given when using personal attributes in decision making, and where these attributes are not used in monitoring for fairness.
3	For non-personal attributes, are there statistical dependencies with personal attributes that cause them to act as material proxies for personal attributes? If so, how are they managed?	As data volumes and the number of potential proxies grows, FSIs should put in place internal standards and guidelines around the identification of material proxies. Non-personal attributes could potentially act as proxies for personal attributes, including for personal data, where statistical dependencies exist, meeting an association criterion as defined by the FSIs. Therefore, the proxies could represent personal attributes, including personal data with all risks, regulatory requirements and processing obligations. The association criterion is an association measure and threshold over which the association is strong enough for the non-personal attribute to be considered a proxy for the associated personal attribute. This measure varies by use case and by FSI.

3.5 Step 3: Build and Validate

3.5.1 Part B(ii): Examine Models

G11 **Is the composition of the AIDA System defined? Is it clear how each component of the system-AIDA models, as well as business rules and human judgement, if relevant - are used to achieve its commercial objectives? Are the performance estimates and the uncertainties of those estimates documented?**

[This question refers to questions B3, B4 and B5 in the Phase 1 Fairness Methodology questions]

Assess Components

The AIDA system often includes more than mathematical models and data, typically incorporating business rules and human judgement. The interactions between these components and their influences on the final AIDA system decisions should be clearly understood and outlined to enable effective performance and fairness checks for system evaluation.

No.	Key considerations	Proposed general guidance *(examples italicised)*
1	What are the separate components of the end-to-end AIDA system?	An AIDA system is composed of different components working together to achieve a commercial objective. The components of an end-to-end AIDA system may include models, upstream business rules (such as eligibility criteria), downstream business rules, human labelling and manual verification.
2	How do these components interact?	The different components of an AIDA system, when arranged in a specific pattern/process flow, interact with each other to serve the business intent. *An example of the process flow of an AIDA system is Model 1 → human labelling → business rules #1 → Model 2 → business rules #2 → Model 3 → Decision*
3	What is the impact of each component on the resulting AIDA system decision?	Each component of the AIDA system has a specific role to play and may impact the decision in different ways. *For example, the upstream business rules may filter out inputs into the system, constraining the input according to a defined criterion.* The system output is a cumulative output of all the components in the system and disproportionate distribution of errors in any component can impact the end outcome.

Assess Models

An AIDA system could consist of multiple models. Each model should be evaluated in its own right, in terms of the performance metrics against target levels and the associated uncertainties. This allows the FSI to assess each model's relative contribution towards the achievement of commercial objectives by the overall AIDA system.

No.	Key considerations	Proposed general guidance *(examples italicised)*
1	What performance metrics are chosen to evaluate each model in the AIDA system and what are the reasons for the choice of metrics?	Ascertaining how well a model is going to perform on unseen data is often quantified through performance metrics. The choice of metrics influences how the performance of a model is measured and contrasted. Choosing the appropriate metric is based on the commercial objective and the nature of data available for training.
2	How are uncertainties in performance metric estimates measured and what the reasons for the choice of this uncertainty measure?	AIDA models learn from historical data and generate predictions for unseen data, with an associated confidence level for the prediction to hold true or that the prediction may be different for other, similar customers. There are methods such as Bootstrap CI and Bayesian CI that can be used to compute uncertainties for performance as well as fairness metrics. Additionally, acceptable methods should be documented in the standard AIDA system development documentation by FSIs. The choice of these method depends on use case (classification/regression), model type, and the data on which the metrics is calculated, among other factors. Refer appendix 8.8 for further details on uncertainty methods.
3	What are the performance metric estimates, and the uncertainty of these estimates, for those at risk of harm identified in F1 relative to the target population groups of the product?	Models may have the same aggregate performance across a validation set but have very different performance for at risk individuals or groups. Calculate values of chosen metrics and associated uncertainties, according to documented methodology, particularly for at risk groups relative to other groups.

Assess Systems

For a meaningful fairness assessment, it is vital to include all AIDA models as well as other non-AIDA components (such as business rules, override conditions and others) which lead to the final decision outcomes from the system. The considerations are similar to the table above, but with the focus now on the overall AIDA system. The responses to this question provide insights into the ability of the overall AIDA system to attain the commercial objectives stated in A1.

No.	Key considerations	Proposed general guidance *(examples italicised)*
1	What performance metrics are chosen to evaluate the AIDA system and what are the reasons for the choice of metrics?	In cases when the AIDA system comprises of only one model, the choice of performance metrics will be that chosen for the model. However, when there is more than one component model in the system, with different performance objectives, then to measure the overall system performance certain component model metrics may be prioritised over other ones used by remaining component models. Clear justifications need to be provided on why a certain performance metric was chosen compared to other ones which other component models may be using. The chosen metric should be carefully aligned with the defined commercial objective.
2	How are the uncertainties in performance metric estimates measured?	Acceptable methods should be documented in the standard AIDA System development documentation by FSIs.
3	Are there aspects of the system's commercial objectives that these performance measures fail to capture, and could this discrepancy hide fairness risks?	Due to various data or design limitations, there may be design decisions/ assumptions made which may not fully translate to the intended commercial objective for the use case. These discrepancies can lie in functional or technical design of the system. *For instance, if the commercial objective is to maximise profits, but due to insufficient information on cost to reach out to customers the system aims to maximise sales instead. Maximising sales will not always mean maximising profits, as the cost may be unevenly distributed across population groups and may be significantly different for certain target groups. Such situation may create systematic disadvantage for groups with low sales but high potential profits.*
4	How are uncertainties in the AIDA system arising from non-AIDA components addressed and minimised?	The non-AIDA components of an AIDA system linked to business rules, human judgement or overrides may have a level of uncertainty and should be assessed. The process for applying business rules and human judgement used in an AIDA system should be clearly documented (including guidance) and understood by the users of the system to reduce user specific variation in outcomes.

3.5.2 Part C: Measure Disadvantage

F9 **Have you assessed and documented the quantitative estimates of the system's performance against its fairness objectives and the uncertainties in those estimates, assessed over the individuals and groups in F1 and the potential harms and benefits in F2?**

[This question refers to question C1 in the Phase 1 Fairness Methodology questions]

The fairness objectives identified in A4 would be translated into quantifiable metrics to enable FSIs to assess the level of fairness achieved by their AIDA systems, where fairness is defined according to G5 (commercial objectives of a system), F1 (groups identified as at risk of disadvantage) and F2 (the harms and benefits experienced by groups at risk of disadvantage). These metrics should be evaluated against target levels that are defined prior to the development of the AIDA system and aligned with internal organisational principles and guidelines on fairness.

No.	Key considerations	Proposed general guidance *(examples italicised)*
1	How is the system's fairness being measured?	Common ways for a system's fairness could be measured are in terms of: • Number of unfair events (volume). • Value (monetary value impact to customers) of unfair events. For instance, if the monetary impacts of benefits or harms differ across different population groups, simply measuring incident rates may not be sufficient and a measure of value for the harm/benefit may need to be estimated for the different population groups to get to a measurement of system fairness. Once it is determined how fairness should be measured, it is then possible to define fairness measures by comparing these rates at which benefit- or harm-causing circumstances occur in the system's operation within cohorts for individual fairness, and across cohorts for group fairness. How to determine an acceptable level (threshold) of difference in rates between groups should be in an approved standard process for selecting and assessing fairness metrics.
2	How are the uncertainties in fairness metric estimates measured?	Computing uncertainty in the fairness measures will depend on the chosen measure, and whether the measure is being evaluated on training data or deployment cohort data. There are several widely applicable and commonly used ways of quantifying uncertainty in estimates, such as bootstrap confidence intervals and Bayesian credible intervals for estimating uncertainty of the fairness measures when evaluating a model's fairness. Refer appendix 8.8 for more information on measuring uncertainties.
3	Do the fairness metrics chosen satisfy group fairness or individual fairness or both?	In addition to satisfying broader fairness objectives at group-level, it is also important if possible to check for individual fairness – i.e., if similar customers have the same decision outcome. Ways that this can be done include through human-review and through local transparency techniques. Ideally FSIs should look for ways to assess both (i.e., compare rates at which benefit or harm causing circumstances occur in the system's operation within cohorts for individual fairness, and across cohorts for group fairness).

F10

Have you assessed and documented the achievable trade-offs between the system's fairness objectives and its commercial objectives?

[This question refers to question C2 in the Phase 1 Fairness Methodology questions]

FSI should aim to fully achieve both commercial and fairness objectives and this is often feasible. However, in some circumstances the objectives may not be fully achieved. The appropriate level of trade-off between competing objectives will rest on a combination of commercial and fairness objectives, internal organisational values, and external factors such as legislation or regulation, societal expectations, and market practice, but also the legitimate interests of individuals and group of individuals including their fundamental rights. Explicitly laying out the trade-off options will allow FSIs to objectively and transparently arrive at an optimal position. A decision on achievable trade-off should be explained and recorded. The trade-offs should be based on a responsible, rational and methodological approach within the state of the art.

No.	Key considerations	Proposed general guidance *(examples italicised)*
1	What mitigation/corrective procedures are implemented to improve a system's measures of fairness for the identified potential disadvantages?	To improve observed (un)fairness of the system, algorithmic in-processing fairness constraints may be implemented in training or in post-hoc corrective procedures (such as calibration) to improve fairness objectives.
2	How is the trade-off analysis carried out and what are its results?	The impact of improving a fairness metric outcome, the quantitative measure of the system's fairness objective, should be assessed against resulting change in the systems commercial metric outcomes, the quantitative measure of the system's commercial objectives. This can be done by comparing the changes in these identified metrics as selection assumptions or procdures of the AIDA system vary (such as decision theshold, choise of models, bias treatments) to select and an optimum balancing point.
3	Which factors are considered in deciding the acceptable *trade-off options*?	Some of the factors that could be considered are: • Business value expectations from the use case. • The organisation's AIDA system performance standards. • Fairness metric acceptable deviation thresholds from similar use-cases or industry standards (if any). An internal standard process for determining trade-offs for consistency across the organisation is recommended. Note that inescapable trade-offs may also exist and reducing one type of systematic disadvantage may increase another type.
4	What aspects of the AIDA design or operations could expose potential trade-offs?	Identify aspects/design assumptions in the system which can allow possible options for potential trade-offs between various system objectives such as decision threshold selection, choice of model algorithm, loss function, hyperparameters, filtering rules, etc.

No.	Key considerations	Proposed general guidance *(examples italicised)*
5	What aspects of the AIDA design or operations were not analysed, and why? Could these affect the potential trade-offs?	In the various factors to take into consideration when determining a trade-off, as identified above, FSIs should indicate operating parameters and design choices which were not analysed as these may relate to partial coverage of target population group, product eligibility criteria which excludes some segments of target group, etc.; and which may cause a differential proportion of benefit or harm across cohorts.
6	What are the minimum acceptable standards for commercial and fairness metrics that form the boundaries of trade-off options?	Internal standards/guidelines for trade-off decisions between fairness and performance are recommended. These standards can include selecting decision frontiers that enable simultaneous achievement reasonable fairness and performance objectives.
7	Are there ways to improve trade-off positions for the system?	Trade-offs can often be improved (in the sense of having to make fewer compromises between objectives) with more accurate predictions, so any approaches to improving predictive accuracy will be beneficial (such as collecting more data, improving data quality, ensembled classifiers and others).

F11

Have you justified (vs alternative tradeoffs) and documented the final fairness outcome for the AIDA System?

[This question refers to question C3 in the Phase 1 Fairness Methodology questions]

Following selection of the optimal, trade-off position on which to operate, the FSI should justify this choice by demonstrating that it provides the most appropriate trade-off between competing objectives for the use case under consideration.

No.	Key considerations	Proposed general guidance *(examples italicised)*
1	What are the justifications for choice of optimal, trade-off position, relative to other trade-off alternatives?	Justifying the choice of fairness metrics and thresholds against system performance trade-offs depend on the relative importance of different objectives and priorities, which the system is looking to address. *One example method is choosing from a "Pareto Frontier", a set of non-dominated solutions, forming an optimal "efficient frontier" for the trade-off between accuracy and fairness (similar to the efficient frontier in portfolio management which quantifies the trade-off between risks and returns). FSIs should determine the optimal point on the frontier with respect to the relative importance between accuracy and fairness. More detail about this method can be found in the book 'The Ethical Algorithm' by Michael Kearns and Aaron Roth, page 81.*

3.6 Deploy and Monitor

3.6.1 Part E: Examine System Monitoring and Review

G12 **Is the system's monitoring and review regime designed to detect abnormal operation?**

[This question refers to question E1 in the Phase 1 Fairness Methodology questions]

Even for an AIDA system that performs well at deployment, anomalies can arise during production that can potentially cause unanticipated harms to individuals or groups. Additionally, for any system to be fair, it first should be performant. Therefore, it is critical to define appropriate KPIs to monitor general performance, as well as relevant measures of fairness and other potential information sources that can help detect unintended harms early on.

No.	Key considerations	Proposed general guidance *(examples italicised)*
1	How, by whom, and how frequently are the system's overall design, objectives, performance, impact and fairness reviewed?	Accountability for the monitoring and review process, including fairness aspects, along with details on how this is to be carried out, should be clearly stated prior to deployment.
2	What KPIs or metrics are tracked post system deployment, to monitor the performance and fairness of the system?	Examples of KPIs include: • *System Performance KPI: precision at a population level.* • *Fairness KPI: precision calculated for at risk population groups and compared to precision at the population level*
3	How does the FSI aim to detect significant movements in key performance and fairness metrics? Are there guidelines that defined what constitutes a significant movement?	It is recommended that breach thresholds to flag significant movements in chosen KPIs (including fairness KPIs) are defined.
4	Are there mechanisms to track how often manual decision override feature is used by FSI decision makers (e.g., a decision maker has the final authority to use/override decision predicted by the system)? Is it being used more for certain population groups that for others?	Manual overrides could result in similar applicants being treated differently. It is recommended that clear override guidelines are defined, and diligent records are kept to help ensure the objectiveness of such decisions. Post-analysis of overrides pooled by at risk population groups can help detect unintended harms due to human bias, with agreed thresholds over which action is taken. A random selection of the decisions impacted by non-AIDA components may be validated by an independent party (which can be internal to FSI).
5	Does the FSI track and analyse customer redressals/complaints and satisfaction with resolution across at risk groups?	Information should be tracked on challenged decisions, where redress was sought, customer complaints and satisfaction with resolutions. This information can aid in conducting pooled reviews of at risk population groups, to identify unintended consequences of the AIDA system's operations.

G13 Is there fallback and/or mitigation plans in place in case of triggers from the system's monitoring and review regime?

[This question refers to question E3 in the Phase 1 Fairness Methodology questions]

Where a review indicates that unintended harms are being created, FSIs should take steps to rectify the impact from these harms. Action plans should also be put in place to prevent repetition of these harms. This could include amendments to the model, withdrawing the model altogether, better enabling users to make informed decisions using the AIDA system or even exploring if customers themselves can contribute towards improvement of the system.

No.	Key considerations	Proposed general guidance *(examples italicised)*
1	Do the guidelines or standards issued by FSIs identify when action is required on an AIDA system, and what that action should be, include fairness considerations?	Guidelines that exist on triggers for action on current AIDA system (such as *retrain or retire the AIDA model in operation*) should include specific fairness related triggers. For example, when a fairness threshold is within x% of being breached, a detailed assessment should be undertaken. Tighter thresholds may be preferred for more high impact use cases (as defined by the FSI).
2	Is adequate training designed for, and provided to, users of the AIDA system for decision support, including potential unfairness scenarios concerning at risk groups?	Training should be delivered to AIDA system users, which could cover topics including understanding of system process, confidence level and uncertainties of outcomes, override provision and potential likely unfairness scenarios.
3	What mechanisms are in place to enable ongoing improvement of the quality and coverage of data inputs and AIDA models to reduce identified or anticipated unfairness scenarios?	*Examples of mechanisms to enable ongoing improvement of data quality and coverage include incentives to brokers and distributors for providing good quality data, sourcing supplementary data, and mechanisms for customers to validate information being used to make decisions and trigger updates if incorrect.*
4	Are there mechanisms in place to enable algorithmic interventions for improving fairness of outcomes?	Depending on the source of bias, there are relevant algorithmic interventions at the pre-processing, in-processing and post processing stages to improve fairness of the AIDA system. Please refer to appendix section 8.5 for more information.

F12 Does the system's monitoring and review regime ensure that the system's impacts are aligned with its commercial and fairness objectives (G5 and F3)?

[This question refers to question E2 in the Phase 1 Fairness Methodology questions]

Over time, business and fairness objectives of the FSI may shift, new model design concepts could emerge, and the legislative, regulatory, social, market or organisational context could change. FSIs should ensure that the monitoring and review process is designed to periodically reassess alignment between an AIDA system's impacts and its fairness objectives, in the face of this dynamic landscape.

No.	Key considerations	Proposed general guidance *(examples italicised)*
1	Are development assumptions/ design decisions validated over time? Do the changes have any implications for at risk groups?	It is recommended AIDA system documentation detailing components, design assumptions and decision process flow of the system is referenced as part of monitoring review, to assess process on the ground against the original intended flow and highlight if changes pose impacts for at risk groups.
2	Are the system's fitness-for-purpose, potential harm(s) and fairness objectives reassessed as new commercial objectives or contexts of the applications emerge?	*An example from the banking sector of a new commercial objective that should trigger a reassessment of fairness objectives would be:* *A decision is made to start using an existing model that currently only auto approves loans to auto decline loans.*
3	Is the choice of personal attributes and associated justifications for use revalidated overtime?	It is recommended AIDA system documentation detailing personal attributes used for decision making, along with justification, as well as those excluded (as per part D of Phase 1 Fairness Methodology) is referenced as part of monitoring review, when a defined trigger for revalidation is met.
4	How are potential changes to data or characteristics of the target population or relationship between input factors and labels to be predicted being monitored, particularly for at risk groups (drift bias)?	To detect drift bias, usual monitoring of statistical properties of data and the model's predictions, should also be run specifically on at risk groups to make sure significant changes for these groups are not missed. In addition, other relevant macroeconomic or regulatory trends should be monitored, for example, has anti-discrimination regulatory requirements, legislation norms or case law surrounding a financial product powered by the AIDA system changed since development?
5	In scenarios where the performance /fairness measures to be monitored cannot be calculated in short term, are there other measures available which can be measured in the short term to aid performance monitoring?	*An example of this scenario in life insurance, as the measure of claims cannot be calculated in the short term. As a result, the health status of a policyholder, measured by a specific medical condition diagnoses could be used as an indicator of mortality risk.*

04 Guidance on Scaling and Embedding the Fairness Methodology across the FSI Organisation

To put the fairness assessment discussed above to practice, FSIs should look to integrate the assessment into their existing risk management, model governance and model development lifecycle processes. This section aims to provide guidance to FSIs for potential ways to incorporate fairness within their existing AIDA system best practices.

4.1 Customising Depth of Fairness Assessment in line with AIDA System Risk

The extent of the fairness assessment described in this Fairness Methodology should be customized to be commensurate to the fairness risk posed by the AIDA system in question. This can be done in two steps. First, determining the risk level of the AIDA system with respect to fairness; and second, using this risk level to determine whether a fairness assessment should be carried out and, if so, the appropriate depth of the assessment.

4.1.1 Determining the Fairness Risk Level of the AIDA System

FSIs typically have digital and data governance frameworks, processes and guidelines already in place. FSIs are encouraged to calibrate those for FEAT assessment based on their own discretion taking into consideration their existing frameworks, materiality of AIDA systems and cost of FEAT assessment and potential mitigation.

FSIs may leverage those existing frameworks to "risk tier" the AIDA system for the FEAT fairness assessment. If using an existing framework, factors specific to assessing fairness risk may need to be added or upweighted. There is guidance in the Veritas Document 3 FEAT Principles Assessment Methodology section 3.2.2. on key factors typically used by FSIs to risk tier AIDA systems, which include:

- Extent of automation in the AIDA driven decision making process
- Complexity of the AIDA model
- Severity and probability of impact on different stakeholders, including affected individuals
- Monetary and financial impact
- Regulatory impact
- Reputational risk
- Use of personal data

Bullet point 3 above, and specifically the impact on affected individuals, should be the most important factor when determining the fairness risk of an AIDA system. Determining the severity and probability of impact on affected individuals. can be informed by the below (some of which are outcomes of questions F1-F2):

- The probability and severity of potential harms to these individuals (e.g., customers)
- Options for recourse
- The number and type of individuals who could be affected by system outcomes
  - This includes assessing the potential for the system to become an industry standard and used by multiple players (in particular when co-developing the AIDA system with third parties; for example, global tech or system providers, regulators and industries association/consortium). Use by multiple players will increase the number and type of individuals who could be impacted

4.1.2 Determining the Application of the Fairness Methodology, Based on the Risk Level of the AIDA System

In line with a risk based approach, FSIs are encouraged to customise the depth of the Fairness Methodology to be commensurate with the risk level of the AIDA system under consideration. Such customisation could relate to the assessment process itself, the level of detail of the assessment, or to the parties involved. FSIs should look to align customisation with their existing practices to work towards firm-wide consistency. This section is similar to section 3.2.3 in the central document, but focuses on customisations for fairness assessment specifically.

Dimensions for customisation of the Fairness Methodology include:

- **Extent of analysis and level of documentation:**
  - For higher risk systems, the assessment may require more extensive analysis and more thorough responses to the questions in the Fairness Methodology, including detailed justification for the choice of fairness metrics, methods of measurement, and included attributes.
  - For systems that may need to include material human intervention, documentation should include training materials that support consistency in human decision making.
  - For lower risk systems, it may be sufficient to provide short summary answers to some or all of the assessment questions.

- **Balancing commercial and fairness objectives:**
  - Higher risk systems may call for greater efforts towards upholding fairness, at the expense of commercial objectives.
  - In such situations, other means of improving the business performance of systems, such as improving data collection volume or quality, may be explored to achieve a more optimal trade-off.
- **Frequency and extensiveness of monitoring regime:**
  - Higher risk systems are likely to require frequent monitoring processes to be put in place following deployment.
  - Risk-appropriate decisions should be made regarding the metrics and outcomes to be monitored, as well as a wider review of the fairness assessment process to ensure previous assessment results to maintain relevance.
- **Review, approval and escalation process:**
  - Higher risk systems may necessitate more stringent review and approval processes, which might be reflected in the number and levels of parties required for sign-off.
  - Breaches in pre-agreed model performance or fairness metric thresholds under higher risk systems could call for greater levels of escalation and intervention to address the matter.

4.2 Beyond Individual Use Cases: Achieving AIDA Fairness at Scale

To scale fairness assessments to AIDA systems across the organisation in a consistent and robust manner, the Fairness Methodology should become part of embedded processes, practices and systems in FSIs. This section outlines recommended areas to consider integrating fairness assessments into the fabric of the FSI and is similar to section 3.2.1 in the Veritas Document 3 FEAT Principles Assessment Methodology document, but focuses in on fairness considerations.

4.2.1 Articulate Fairness Principles

Articulate a proactive, consistent message on the FSI's approach to fairness in the use of data and algorithms, and its impact on customers and employees.

4.2.2 Embed Fairness Considerations in Existing Risk Frameworks and Governance Structures

Creating a new, dedicated framework for AIDA fairness (or for FEAT more broadly) will be unnecessary in most cases. Instead, FSIs should focus on enhancing existing risk frameworks to meet the requirements around fairness. For example, existing frameworks such as those around model risk (or its equivalent outside banking), data privacy and the fair treatment of customers could be enhanced. Additionally, FSI should update relevant transparent, cross domain governance structures to include fairness, including identifying roles, expectations and accountability.

4.2.3 Define Standards and Processes for Fairness in Key Areas

To enable robust, consistent fairness assessments across the organisation, it is important to have defined standards and processes within an internal fairness framework approved via the relevant governance structures. These standards and processes should cover areas such as identifying at risk groups; personal attributes; proxies; explicit fairness objectives and associated measures and thresholds; and technical documents cover toolkits of algorithmic methods to identify and reduce unfairness. This is covered in more detail in section 3.2.

4.2.4 Appropriate Training

Appropriate training on AIDA Fairness of employees at all levels of the organisation including management, business and risk/control functions, as well as partners and even customers should be undertaken. The more fairness can be understood – in its complexities and necessary trade-offs and in the organisation's principles and implemented standards in the area – the more fairness becomes second nature to employees. Training should also include best practices in understanding the sources of bias and how to prevent them (e.g., data selection issues, common human cognitive biases, decision fatigue when making large volumes of data or engineering decisions, etc.).

4.2.5 Diverse Teams

Actively encourage diversity on the teams designing, developing, approving, deploying and monitoring AIDA Systems to ensure there is input from varied perspectives and backgrounds and to avoid unconscious bias and group think.

05 Topics for Future Consideration

As FSIs and fintech companies drive the development and adoption of AI with the intention of improved financial inclusion, and with governments playing a critical role in ensuring benefits are delivered and harms reduced across society, all need to share a common objective that AI becomes a tool for societal good and does not exacerbate existing socio-economic challenges. In creating this document, the Veritas Fairness workstream has identified areas where greater clarity would be beneficial.

- **Alignment between AIDA and non-AIDA fairness requirements.** It will be important to align AIDA fairness with existing compliance frameworks (e.g., Treating Customers Fairly principles).
- **The treatment of AIDA usage via third parties.** FEAT guidelines, as well as those from other regulators, are clear in holding FSIs accountable for AIDA fairness (and transparency/ethics). However, when leveraging a third party for an AIDA score or insight, the FSI may not always have full visibility into the vendor's use of AIDA, particularly because of vendors' concerns around sharing intellectual property. Guidance on how to achieve alignment on fairness objectives and validation of the assessment between all parties involved in the AIDA system development throughout the AIDA system development cycle would be beneficial.
- **Acceptable standards of fairness for high risk and low risk use cases in FSIs.** As fairness is context and application dependent, domain-specific educational resources, processes, and procedures to guide FSIs would be beneficial. An area of exploration could be to identify the main use cases used at scale by FSIs (or those with good potential to be), along with associated risk tiers, and to develop an integrated FEAT principles assessment methodology with guidance on how to scale and case studies to provide practical guidance.
- **Set of procedures to check and correct for biases in data and algorithms.** Practitioners have highlighted concerns over the time and effort required to custom fit assessments to use cases, and the limitations of existing methods in the literature for surfacing fairness issues in complex, multicomponent AIDA systems, where fairness can be heavily context dependent, and the space of potential contexts is often very large. Similar concerns arise around determining whether specific issues (e.g., complaints from customers) are "one-offs" or symptomatic of broader, systemic problems that might require deeper investigation. Developing procedures for automated auditing processes/tools and refined approaches tailored to various types of AIDA systems would help to address these concerns.

- **Fairness by unawareness of personal data.** Fairness assessment methods often assume access to demographics at the individual level, but many teams are only able to collect such information at coarser levels, if at all, as personal data usage/ access is restricted by law and regulation. Future research should explore ways either to support fairness auditing with access to only coarse-grained demographic information (e.g., neighbourhood level demographics) or with no access at all, when they might use privacy preserving methods to conduct fairness assessments (e.g., homomorphic encryption, differential privacy, synthetic data generation)
- **Extending fairness beyond individual customers to SMEs/corporates.** This approach could be particularly relevant when promoters', directors', experts' or investors' personal details are used in business-to-business (B2B) use case applications and therefore also potentially in AIDA systems.
- **Alignment on what constitutes "consumer consent" on usage of personal attributes, including personal data, for AIDA fairness assessment.** The future regulatory environment should provide adequate protection of data privacy for individuals while allowing AIDA system owners to access lawfully the necessary data allowing them to assess fairness for relevant population subgroups to limit discrimination of the designed AIDA systems and at the same time safeguarding privacy of the individuals.
- **Better quality datasets.** Future work should also support practitioners in collecting and curating high quality datasets, with an eye towards fairness in downstream models.
- **Cross-border fairness assessment:** Despite AIDA system fairness assessment usually being jurisdiction bounded, there is value for global FSIs to develop tools and frameworks to adopt best practices, measure fairness and reduce discrimination at a global scale. One of the challenges to adopt such global AIDA fairness practice is linked to cross-border data transfer requirements applicable when sharing data across jurisdictions. Transfer learning techniques and adoption of synthetic data could support this objective. However future work should clarify and address this challenge.

06 Acknowledgements

Role	Name	Organisation
Lead Subject Matter Expert	Medb Corcoran	Accenture
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Data Scientist	Visheshta Khanna	Accenture
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Lead Academic	Adrian Weller	Alan Turing Institute
Lead Subject Matter Expert	Yannick Even	Swiss Re
Project Coordinator & Risk Management	Jasmine Ong	Swiss Re
Project Coordinator & Data Scientist	Charlene Lee	Swiss Re
Lead Data Scientist	Luca Baldassarre	Swiss Re
L&H Underwriting Expert	Farooque Ahmed	Swiss Re
L&H Underwriting Expert	Jon Lambert	Swiss Re
Risk Management Expert	Lutz Wilhelmy	Swiss Re
Legal & Compliance Expert	Tomas Abelovsky	Swiss Re

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08 Appendix

Below technical appendices 8.1 - 8.10 are included to provide additional background reading for some of the key technical concepts found in the Fairness Methodology document. **They are not exhaustive and represent a sample at a point of time in an area where new developments are being published regularly.**

8.1 Measuring Fairness of Decision Outcomes (Group & Individual Fairness metrics)

The list of metrics in this section measure group fairness and cover the main fairness metrics mentioned in Phase I and Phase II Fairness Methodology and use case documents. It is important to note that the metrics apply to a binary classification problem.

In a non-binary or multi-classification situation where the number of classes to be predicted are larger than two, the fairness metrics typical used in a binary setting can also be employed – see the last paper in the references at the bottom of this subsection for details. Parity is achieved for a fairness metric if the rates are the same across subgroups of the personal attribute. The metrics can be applied in a fairness context as ratios that measure percentage differences between subgroups (a common approach) in which case parity is 1, or as absolute differences between the subgroups – in which case parity is 0. For example, for the fairness metric of Positive Predictive Rate (first in the table below) applied to Gender Marital Status:

1. It could be looked at as a ratio: Disparate Impact:
   - % not married females with positive predictions/% married with positive predictions
   - In which case the closer the ratio is to 1 (parity), the more balanced it is
2. Or as a difference: Demographic Parity:
   - % not married with positive predictions - % married with positive predictions
   - In which case the closer the difference is to 0 (parity), the more balanced it is

Confusion Matrix (for formula):

		Predicted Class: Positive	Predicted Class: Negative
Actual Class	Positive	TP	FN
Actual Class	Negative	FP	TN

Metric	Definition	Formula (in terms of TP, TN, FP, FN)
Positive Predictive Rate	The fraction of positive predictions out of all predictions. When the metric is applied in a fairness context as a ratio, it is also called **Disparate Impact**. When applied in a fairness context as absolute difference, it is called **Demographic Parity or Proportional Parity.**	(TP + FP) / (TP + FP + TN + FN)
False Negative Rate	The fraction of positive cases which were incorrectly predicted to be negative out of all the positive cases. It represents the probability of the positive subjects to be classified incorrectly as negative ones.	FN / (TP + FN)
False Omission Rate	The fraction of negative predictions which were actually positive out of all the negative predictions. It represents the probability of an erroneous negative prediction.	FN / (TN + FN)
True Positive Rate	The fraction of positive cases which were correctly predicted to be positive out of all the positive cases. It represents the probability of the positive subjects to be classified correctly as positive ones. It is also called recall. When this metric is applied in a fairness context as a ratio or difference, it is also called **Equal Opportunity.**	TP / (TP + FN)
Positive Predictive Value	The fraction of positive cases which were correctly predicted out of all the positive predictions. It is usually referred to as precision and represents the probability of a correct positive prediction. When this metric is applied in a fairness context as a ratio or difference, it is also called **Predictive Value Parity.**	TP / (TP + FP)
False Positive Rate	The fraction of negative cases which were incorrectly predicted to be positive out of all the negative cases. It represents the probability of the negative subjects to be classified incorrectly as positive ones.	FP / (TN + FP)
False Discovery Rate	The fraction of positive predictions which were actually negative out of all the positive predictions. It represents the probability of an erroneous positive prediction.	FP / (FP + TP)
Average Odds	The average of true positive rate and false positive rate. When this metric is applied in a fairness context as a ratio or difference, it is also called **Equalised Odds**.	(TP / (TP + FN) + FP / (TN + FP))/2
Prevalence Rate	The fraction of that population that are positive cases (had the positive actual outcome) out of the whole population. When this metric is applied as a ratio it is called **Prevalence Ratio**. This metric is applied in a fairness context to the ground truth data rather than model outcomes to determine if there is an inherent balance (or imbalance) in terms of the target (positive) outcome in personal attribute subgroups in the historic population.	(TP + FN) / (FP + TP + TN + FN)

References

- https://fairware.cs.umass.edu/papers/Verma.pdf - Fairness Definition Explained
- https://en.wikipedia.org/wiki/Fairness_%20-%20machine_learning - Fairness definitions
- https://arxiv.org/abs/2003.00827
- https://www-axa-com.cdn.axa-contento-118412.eu/www-axa-com/d6324958-367e-4375-81c3-cfeb8e7ccc66_AXA_FairnessCompass-English.pdf - Fairness Compass definitions explained

In the literature, **individual fairness** tends to mean that similar individuals are treated similarly. What constitutes similarity between individuals is an open question and is usually left to the determination of a domain expert for the use case. What it means to be treated similarly is also more subtle when talking about individuals, since similar risk scores are more likely to lead to different outcomes when making comparisons at an individual level. At this time there are no set of commonly used individual fairness metrics. There is a paper in the next section that documents a potential approach.

8.2 Selecting the Most Appropriate Fairness Metric

Fairness is closely related to the concept of justice and standard of rightness. Various definitions exist, some of them conflicting with each other. In the absence of a uniformly accepted notion of fairness, choosing the right kind for a specific situation is a non-trivial task. When it comes to assessing fairness in AIDA systems, this task becomes: How to select the most appropriate fairness metric for a particular application. Several toolkits/frameworks exist, such as the fairness metrics tree which is illustrated in the Veritas Phase 2 - Illustrative Case Studies document under question A4, for selecting the most appropriate fairness metrics for a specific use case application.

The references below give details on some of these frameworks. It is up to each FSI to determine the framework for their circumstances.

Conceptual References

Group Fairness

- https://www-axa-com.cdn.axa-contento-118412.eu/www-axa-com/d6324958-367e-4375-81c3-cfeb8e7ccc66_AXA_FairnessCompass-English.pdf - A fairness compass (decision tree) to select the most appropriate fairness metric
- https://arxiv.org/pdf/2001.07864.pdf - Study of the most widely used fairness metrics using a common mathematical framework. The study also presents new results on the relationships among the different fairness metrics.
- https://arxiv.org/pdf/1811.05577.pdf - Aequitas, an open source bias and fairness audit toolkit
- https://arxiv.org/abs/1511.00148 - A survey to review and organise various discrimination measures that have been used for measuring discrimination in data, as well as in evaluating performance of discrimination aware predictive models.
- https://doi.org/10.1145/3351095.3372864 - A comparative study for selecting group and individual fairness metrics.
- https://arxiv.org/abs/1808.00023 - A paper which presents material on the limitations of common formulations of fairness

Individual Fairness:

- https://arxiv.org/pdf/1104.3913.pdf - This paper is a framework for fair classification comprising (1) a (hypothetical) task-specific metric for determining the degree to which individuals are similar with respect to the classification task at hand; (2) an algorithm for maximising utility subject to the fairness constraint, that similar individuals are treated similarly.

Technical Code References

- https://biasfix.com/select-fairness-metrics/
- https://github.com/Trusted-AI/AIF360
- https://github.com/dssg/aequitas
- https://axa-rev-research.github.io/fairness-compass.html

8.3 Deciding Deviation Threshold from Fairness Criteria

When any of the above fairness measures deviate from their fairness criteria, a choice needs to be made as to what threshold(s) should be used to determine if further action should be taken. There is currently little regulation around what these thresholds should be. The 4/5th rule, or 20% deviation from ideal fairness metric value, a method sometimes used today, is based on an approach taken in the US, which a number of US government agencies adopted from the Uniform Guidelines for Employee Selection Procedures.

The references in the table below explain more about this – but it is up to each FSI to determine the threshold for their circumstances.

Conceptual References	Technical Code References
• https://fairmlbook.org/tutorial1.html - A tutorial on disparate impact and the 4/5th rule. • https://www.eeoc.gov/laws/guidance/questions-and-answers-clarify-and-provide-common-interpretation-uniform-guidelines - Background on where the 4/5 rule came from and has been interpreted over time • https://www.wikiwand.com/en/Disparate_impact#/Adverse_impact - Introduction to adverse impact and 80% rule. • https://www.hirevue.com/blog/hiring/what-is-adverse-impact-and-why-measuring-it-matters - Introduction to adverse impact and methods to measure it (four fifth rule)	• https://github.com/visionjo/facerec-bias-bfw

8.4 Types of Algorithmic Biases

Algorithmic bias describes systematic and repeatable errors in a computer system that create unfair outcomes, such as privileging one arbitrary group of users over others. Bias can emerge due to many factors, including but not limited to the design of the algorithm (model bias) or the unintended or unanticipated use or decisions relating to the way data is coded, collected, selected or used to train the algorithm (data bias).

The references in the table below give more detail on different types of bias as well as potential impacts.

Conceptual References	Technical Code References
• https://arxiv.org/abs/1901.10002 - A framework that identifies seven distinct potential sources of downstream harm in machine learning, spanning data collection, development, and deployment. • https://arxiv.org/pdf/1908.09635.pdf - A survey that investigates different real world applications that have shown biases in various ways and that lists different sources of biases that can affect AI applications. • http://conflict.lshtm.ac.uk/page_39.htm - Critical concepts related to bias and errors. • https://quantifyinghealth.com/proxy-bias/ - Introduction to proxy bias with examples • http://proceedings.mlr.press/v108/fogliato20a.html- When target variable bias is a concern, the sensitivity analysis framework presented in this paper can be used to quantify the level of bias sufficient to call into question conclusions about the fairness of a model obtained from biased observed data. • https://humanrights.gov.au/about/news/media-releases/infographic-historical-bias-ai-systems - Introduction to historical bias and mitigation techniques with example.	• https://github.com/Trusted-AI/AIF360 • https://github.com/ropensci-archive/proxy-bias-vignette • https://github.com/nikolaydubina/go-featureprocessing • https://keras.io/keras_tuner/ • https://github.com/intuit/bias-detector • https://github.com/Call-for-Code-for-Racial-Justice/bias-detection-engine • https://github.com/ai-med/Dataset-Bias • https://github.com/ModelOriented/fairmodels • https://github.com/dssg/aequitas



8.5 Technical Bias Mitigation Methods

The potential mitigation actions for debiasing an AIDA system can be as diverse as the use cases for AIDA systems and depend on the specific findings of a fairness assessment. Mitigation can include interventions like bias avoidance at source (e.g., data collection enhancements). The mitigation methods focused on here are those that can be implemented during the AIDA system development. There are three main intervention points that FSI or their AIDA System Developers may consider:

- **Pre-processing.** This occurs before the creation of the model. Using these techniques, the dataset is transformed with the aim of removing the underlying bias from the data before modelling.
- **In-processing.** In-processing techniques can be considered as modifications to the traditional learning algorithms to address discrimination during the model training phase.
- **Post-processing.** In post-processing techniques, the processing of the dataset is done after model training. For example, adversarial debiasing.[25]

The references in the table below give more detail on some bias mitigation techniques.

Technique Type	Conceptual References	Technical Code References
Pre-processing Techniques	• https://www.cs.toronto.edu/~toni/Papers/icml-final.pdf - The paper describes a pre-processing approach to fairness that consists in an optimisation problem of finding a good representation of the data with two competing goals: to encode the data as well as possible, while simultaneously obfuscating any information about membership in the protected group.	AIF360: Learning Fair Representation • https://github.com/Trusted-AI/AIF360
	• https://arxiv.org/pdf/1904.07911.pdf - Representation bias Removal (REPAIR) method to reduce the representation biases of a dataset.	Dataset REPresentAtion bIas Removal (REPAIR) • https://github.com/JerryYLi/Dataset-REPAIR
In-processing Techniques	• https://arxiv.org/pdf/1801.07593.pdf - A framework for mitigating biases by including a variable for the group of interest and simultaneously learning a predictor and an adversary. The input to the network X, produces a prediction Y, while the adversary tries to model a protected variable Z.	AIF360: Adversarial Debiasing • https://aif360.readthedocs.io/en/latest/modules/generated/aif360.algorithms.inprocessing.AdversarialDebiasing.html#aif360.algorithms.inprocessing.AdversarialDebiasing
	• https://arxiv.org/pdf/1803.02453.pdf - A systematic approach for achieving fairness in a binary classification setting. The key idea is to reduce fair classification to a sequence of cost sensitive classification problems, whose solutions yield a randomised classifier with the lowest (empirical) error subject to the desired constraints.	Fairlearn: Reductions GridSearch • https://fairlearn.org/main/api_reference/fairlearn.reductions.html#fairlearn.reductions.GridSearch
	• https://arxiv.org/pdf/2006.05109.pdf - A general constrained Bayesian optimisation (BO) framework to optimise the performance of any ML model while enforcing one or multiple fairness constraints. BO is a model-agnostic optimisation method that has been successfully applied to automatically tune the hyperparameters of ML models.	AutoGluon: Constrained Hyperparameter Optimization • https://github.com/awslabs/autogluon/
	• https://arxiv.org/pdf/1706.02409.pdf - A flexible family of fairness regularisers for (linear and logistic) regression problems.	Fair-regression • https://github.com/Azizmj/Fair-regression
Post-processing Techniques	• https://arxiv.org/pdf/1709.02012.pdf - This paper presents a post-processing technique that optimises over calibrated score outputs to find probabilities with which to change output labels with an equalised odds objective.	Equalised Odds and Calibration • https://github.com/gpleiss/equalized_odds_and_calibration?utm_source=catalyzex.com
	• https://pubs.acs.org/doi/10.1021/acs.jcim.1c00160# - Introduction to GHOST: a method for adjusting the decision threshold to handle imbalanced data in machine learning	GHOST • https://github.com/rinikerlab/GHOST
	• https://arxiv.org/abs/1610.02413v1 - Hardt, M., Price, E., & Srebro, N. (2016). Equality of opportunity in supervised learning. Advances in Neural Information Processing	Constrained Balanced Accuracy • https://github.com/veritas-project/phase1/tree/main/credit_scoring

8.6 Drift bias

Model drift[26] is the degradation of a model's prediction power due to changes in the environment, and specifically changes in the relationship between the target variable and the independent variables changes over time. Model drift can be classified into two broad categories. The first type is called **'concept drift'**. This happens when the statistical properties of the target variable itself change. The second and the more common type is **'data drift'**. This happens when the statistical properties (e.g., distribution) of the predictors change.[27] These types of drift can also cause unintentional bias for subgroups of the population that the model is applied to.

The references in the table below give more detail on detecting and addressing drift bias.

Conceptual References	Technical Code References
• https://arxiv.org/pdf/2012.09258.pdf - Detection of data drift and outliers affecting model performance over time. Detecting drift indirectly by nonparametrically testing the distribution of model prediction confidence for changes. • https://www.cs.upc.edu/~abifet/EDDM.pdf - A method for detecting concept drift, even in the case of slow gradual change. It is based on the estimated distribution of the distances between classification errors. • https://www.mdpi.com/2306-5729/6/7/77/htm - This work focuses on addressing data randomness and concept drift by minimising modelling variations and yielding consistent results across samples.	• https://github.com/alipsgh/tornado • https://github.com/anodot/MLWatcher/blob/master/readme.md • https://github.com/MicrosoftDocs/azure-docs/blob/master/articles/machine-learning/how-to-monitor-datasets.md • https://github.com/evidentlyai/evidently • https://github.com/ModelOriented/drifter • https://github.com/SeldonIO/alibi-detect

8.7 Fairness-Performance Trade-offs

The fairness-performance trade-off is an important concept which states that when some notion of fairness is enforced then often the performance of the AIDA system suffers. This depends on the fairness metrics, performance metrics and the dataset used. There are number of processes available for carrying out trade-off analysis.

The references in the table below give more detail on fairness-performance trade-offs.

Conceptual References

- http://proceedings.mlr.press/v81/menon18a/menon18a.pdf - A deep dive into the trade-offs between accuracy for a given level of fairness in terms of costs, as well as establishing an efficient set (Pareto set) of classifier threshold settings.
- https://arxiv.org/pdf/1910.07870.pdf - This paper examines fair classification through the lens of mismatched hypothesis testing: trying to find a classifier that distinguishes between two ideal distributions when given two mismatched distributions that are biased. Using Chernoff information, a tool in information theory, it theoretically demonstrates that, contrary to popular belief, there always exist ideal distributions such that optimal fairness and accuracy (with respect to the ideal distributions) are achieved simultaneously: there is no trade-off.
- https://arxiv.org/pdf/2004.034.pdf - This paper frames several optimisation problems that directly optimise both accuracy and fairness objectives, which yield a general perspective for understanding multiple trade-offs including group fairness incompatibilities.
- https://arxiv.org/pdf/2008.10797.pdf - The experiments in this paper reveal that for training data already exhibiting disparities, a newly introduced causal notion of fairness may be capable of traversing a greater part of the fairness-performance space, relative to more standard measures such as demographic parity and conditional parity. The experiments also reveal that tools from multi-objective optimisation are crucial in efficiently estimating the Pareto front (i.e., by finding more non-dominated points), relative to other sensible but ad-hoc approaches.
- https://www.researchgate.net/publication/345555273_There_is_no_trade-off_enforcing_fairness_can_improve_accuracy - This paper shows that trade-off may not be necessary. If the algorithmic biases in an ML model are due to sampling biases in the training data, then enforcing algorithmic fairness may improve the performance of the ML model on unbiased test data.
- https://arxiv.org/pdf/2106.02705.pdf - This presents a multi-dimensional Pareto frontier on (1) the trade-off between group fairness and accuracy with respect to each task, as well as (2) the trade-offs across multiple tasks.

Technical Code References

- https://github.com/google-research/tensorflow_constrained_optimization
- https://github.com/visionjo/facerec-bias-bfw
- https://github.com/sul217/MOO_Fairness
- https://github.com/dssg/peeps-chili
- https://github.com/Trusted-AI/AIF360

8.8 Measuring Uncertainties in Fairness/Performance Estimates

AIDA systems, particularly those using machine learning or statistical models, require performance measures (including those relating to fairness) to be quantified mathematically so that they can be optimised or constrained. However, these measures are necessarily computed from finite samples of data, or may even be estimates from other models, and so have an inherent statistical sampling and estimation uncertainty. Furthermore, some of the sample sizes may be very small, particularly for measures of fairness where underrepresented demographic groups make up the sample. Since consequential decisions will be made based upon the values of these measures, it is also important to convey the uncertainty inherent in their values. Bootstrapping was used in the Phase 2 Fairness Use Case to estimate uncertainties for performance and fairness.

The references in the table below give more detail on measuring uncertainties.

Conceptual References

- https://arxiv.org/pdf/1910.09457.pdf - This paper provides an introduction to the topic of uncertainty in machine learning as well as an overview of attempts so far at handling uncertainty in general and formalising this distinction in particular.
- https://arxiv.org/pdf/1907.06890.pdf - This literature proposes a novel framework for uncertainty estimation based on Bayesian belief networks and Monte-Carlo sampling. It fully models the different sources of prediction uncertainty and incorporates prior data information
- https://link.springer.com/chapter/10.1007/978-3-030-44584-3_18 - This is one of the first deep learning studies which quantifies uncertainty and model interpretability in multilabel classification
- https://arxiv.org/pdf/1809.07653.pdf - This presents a scheme to obtain an inexpensive and reliable estimate of the uncertainty associated with the predictions of a machine learning. The scheme is based on resampling, with multiple models being generated based on subsampling of the same training data.
- https://doi.org/10.1214/ss/1032280214 – Introduction to bootstrapping to calculate uncertainties in estimates.

Technical Code References

- https://github.com/uzh-rpg/deep_uncertainty_estimation?utm_source=catalyzex.com
- https://github.com/uncertainty-toolbox/uncertainty-toolbox
- https://github.com/google/TrustScore
- https://github.com/IBM/UQ360?utm_source=catalyzex.com



8.9 AIDA system documentation examples

AIDA Systems should have robust documentation detailing key steps and decisions taken in the end-to-end lifecycle. In addition to having documentation for the end-to-end system, it is especially important to have sufficient detail when documenting the data and model(s) – this is leveraged when doing a fairness assessment. If a FEAT assessment is completed (including Fairness), it is recommended for the assessment to be held with the existing standard AIDA system documentation in order to manage version control.

The references in the table below give some example approaches to AIDA System documentation.

Conceptual References	Technical Code References
• https://arxiv.org/pdf/1810.03993.pdf - This paper proposes a framework called model cards, to encourage transparent model reporting. Model cards are short documents accompanying trained machine learning models that provide benchmarked evaluation in a variety of conditions, such as across different cultural, demographic, or phenotypic groups (e.g., race, geographic location, sex, Fitzpatrick skin type) and intersectional groups (e.g., age and race, or sex and Fitzpatrick skin type) that are relevant to the intended application domains. • https://arxiv.org/pdf/2009.11190.pdf - The enterprise AI canvas is designed to bring data scientist and business expert together to discuss and define all relevant aspects which need to be clarified in order to integrate AI based systems into a digital enterprise. • https://www.microsoft.com/en-us/research/uploads/prod/2019/01/1803.09010.pdf - Datasheets for dataset recommends that every dataset be accompanied with a datasheet documenting its creation, composition, intended uses, maintenance, and other properties.	N/A

8.10 Privacy preserving techniques

AIDA systems could rely on data that is private and sensitive. Privacy enhancing technologies (PET) are technologies that embody fundamental data protection principles by minimising personal data use and maximising data security while enabling appropriate data processing. PETs use techniques to minimise possession of personal data without losing the functionality of an information system. Within PETs, Privacy preserving techniques (PPT) describes mechanisms to perform computations on data while keeping the data private.[28] Synthetic data generation was used in the Phase 2 Fairness Use Case to keep the client data private.

The references in the table below give detail on some PPTs.

Conceptual References

- https://www.ijstr.org/final-print/mar2017/A-Review-Of-Synthetic-Data-Generation-Methods-For-Privacy-Preserving-Data-Publishing.pdf - In this paper different fully and partially synthetic data generation techniques are reviewed, and key research gaps are identified which needs to be focused in the future research
- https://arxiv.org/ftp/arxiv/papers/1804/1804.11238.pdf - This article aims to provide an introduction to the intersection of machine learning and privacy communities fields with special emphasis on the techniques used to protect the data.
- https://www.cis.upenn.edu/~aaroth/Papers/privacybook.pdf - After motivating and discussing the meaning of differential privacy, the preponderance of this monograph is devoted to fundamental techniques for achieving differential privacy, and application of these techniques in creative combinations, using the query release problem as an ongoing example.
- https://www.researchgate.net/publication/328134142_Generation_of_Synthetic_Data_with_Generative_Adversarial_Networks - This thesis focuses on one of the most interesting and promising innovations of the last years in the machine learning community: Generative Adversarial Networks. An approach for generating discrete, continuous or text synthetic data with Generative Adversarial Networks is proposed, tested, evaluated and compared with a baseline approach.
- https://arxiv.org/pdf/2104.11797.pdf - This work investigates the use of synthetic images, created by generative adversarial networks (GANs), as the only source of training data.

Technical Code References

- https://github.com/LatticeX-Foundation/Rosetta
- https://github.com/innovation-cat/Awesome-Federated-Machine-Learning
- https://github.com/dilawarm/federated
- https://github.com/sdv-dev
- https://github.com/tirthajyoti/Synthetic-data-gen
- https://github.com/ydataai/ydata-synthetic
- https://github.com/capitalone/synthetic-data
- https://github.com/theodi/synthetic-data-tutorial

8.11 Fairness Concepts Specific to Insurance

Fairness concepts specific to insurance include:

- **Community and Individual Pricing**

  Insurance companies set premium levels to cover risks assumed from a diverse group of customers. The two ends of the pricing strategy continuum are:

  i. Community pricing, where every customer is charged the same; or
  ii. Individual pricing, where each customer is charged according to their unique characteristics relative to the pool.

  While often referred to as being at opposite ends of the spectrum, it is possible to also explore combinations, such as where community pricing is applied for the base cover and individual pricing applied on optional top-ups.[29]

  For instance, in markets such as Australia required life cover can be provided through mandatory superannuation arrangements, by way of group insurance schemes. However, consumers may also purchase more cover in a private insurance environment.

- **Actuarial and behavioural fairness**

  Actuarial fairness is based on the concept that individuals bearing similar levels of risk are charged similar premiums and has been traditionally used in the private insurance field. Behavioural fairness moves away from this conventional practice towards the realm of personal responsibility, where rewards are offered, or penalties imposed based on behavioural choices within the control of individuals. As to which is a "fairer" concept, this is still a matter of debate.[30]

  Notwithstanding the differences, it is acknowledged that for both banking and insurance, cross functional discussions among management, data scientists, and experts in the fields of law, ethics and compliance are critical to determining the optimal strategy for fairness in their AIDA systems.





8.12 Examples of Non-Binding AIDA Principles

- The Principles published by MAS that was released in 2018. The Principles that explicitly relate to fairness (see section 1.2.1 above) aim to ensure that AIDA driven decisions do not disadvantage any individual or groups of individuals, without appropriate justification of inputs, processes and outcomes.
- Additionally, principles outlined in the Fair Dealing Outcomes that were also published by MAS put customers' interest at the centre of the FSI's corporate culture.
- In June 2021, the European Insurance and Occupational Pensions Authority (EIOPA) published a report from its consultative expert group on digital ethics, which set out artificial intelligence governance principles for an ethical and trustworthy AI in the European insurance sector.[31]
- In April 2021, the European Commission published a draft regulation laying down harmonised rules on AI through the Artificial Intelligence Act with the aim of ensuring a high-level protection of health, safety and fundamental rights in the EU, in addition to increasing trust in the development and uptake of AI.
- In July 2021, the US State of Colorado enacted legislation to tackle issues of unfair discrimination and data and algorithmic bias – an act to restrict insurers' use of external consumer data.[32]
- At the international level, in May 2019, the Organisation for Economic Cooperation and Development (OECD) adopted principles to promote artificial intelligence that is innovative and trustworthy, which respects human rights and democratic values.

8.13 Mapping of Veritas Phase 1 Fairness Methodology Questions to Veritas Phase 2 FEAT Checklist Questions

Question Number	Veritas Phase 1 Fairness Methodology Questions	FEAT Checklist Question Number	Veritas Phase 2 FEAT Checklist Questions
A1	What are the business objectives of the system and how is AIDA used to achieve these objectives?	**G5**	Have you documented the commercial objectives of the system and the quantitative measures to meet those commercial objectives? Is it documented how is AIDA used to achieve them?
A2	Who are the individuals and groups that are considered to be at risk of being systematically disadvantaged by the system?	**F1**	Have you identified and documented who are the individuals and groups that are considered to be at risk of being systematically disadvantaged by the system?
A3	What are potential harms and benefits created by the system's operation that are relevant to the risk of systematically disadvantaging the individuals and groups in A2?	**F2**	Have you identified and documented the potential harms and benefits created by the system's operation that are relevant to the risk of systematically disadvantaging the individuals and groups in F1?
A4	What are the fairness objectives of the system, with respect to the individuals and groups in A2 and the harms and benefits in A3?	**F3**	Have you identified and documented the fairness objectives of the system and associated fairness metrics, with respect to the individuals and groups in F1 and the harms and benefits in F2?
B1	What errors, biases or properties are present in the data used by the system that may impact the system's fairness?	**F4**	Have you documented key errors, biases or properties present in the data used by the system that may impact the system's fairness?
B2	How are these impacts being mitigated?	**F5**	Have you documented how are these impacts being mitigated?
B3	How does the system use AIDA models (with, or separately from, business rules and human judgement) to achieve its objectives?	**G11**	Is the composition of the AIDA System defined? Is it clear how each component of the system-AIDA models, as well as business rules and human judgement, if relevant - are used to achieve its commercial objectives? Are the performance estimates and the uncertainties of those estimates documented?
B4	What are the performance estimates of the AIDA models in the system and the uncertainties in those estimates?		
B5	What are the quantitative estimates of the system's performance against its business objectives and the uncertainties in those estimates?		

Question Number	Veritas Phase 1 Fairness Methodology Questions	FEAT Checklist Question Number	Veritas Phase 2 FEAT Checklist Questions
C1	What are the quantitative estimates of the system's performance against its fairness objectives and the uncertainties in those estimates, assessed over the individuals and groups in A2 and the potential harms and benefits in A3?	**F9**	Have you assessed and documented the quantitative estimates of the system's performance against its fairness objectives and the uncertainties in those estimates, assessed over the individuals and groups in F1 and the potential harms and benefits in F2?
C2	What are the achievable trade-offs between the system's fairness objectives and its other objectives?	**F10**	Have you assessed and documented the achievable trade-offs between the system's fairness objectives and its commercial objectives?
C3	Why are the fairness outcomes observed in the system preferable to these alternative trade-offs?	**F11**	Have you justified (vs alternative tradeoffs) and documented the final fairness outcome for the AIDA System?
D1	What personal attributes are used as part of the operation or assessment of the system?	**F6**	Have you determined and documented personal attributes that are used as part of the operation or fairness assessment of the system?
D2	How did the process of identifying personal attributes take into account ethical objectives of the system, and the people identified as being at risk of disadvantage?	**F7**	Does the process of identifying personal attributes take into account ethical objectives of the system, and the people identified as being at risk of disadvantage?
D3	For every personal attribute and potential proxy for a personal attribute, why is its inclusion justified given the system objectives, the data, and the quantified performance and fairness measures?	**F8**	Have you assessed and documented for every personal attribute and potential proxy for a personal attribute, why is its inclusion justified given the system objectives, the data, and the quantified performance and fairness measures?
E1	How is the system's monitoring and review regime designed to detect abnormal operation and unintended harms to individuals or groups?	**G12**	Is the system's monitoring and review regime designed to detect abnormal operation?
E2	How does the system's monitoring and review regime ensure that the system's impacts are aligned with its fairness and other objectives (A1 and A4)?	**F12**	Does the system's monitoring and review regime ensure that the system's impacts are aligned with its commercial and fairness objectives (G5 and F3)?
E3	What are the mechanisms for mitigating unintended harms to individuals or groups arising from the system's operation?	**G13**	Is there fallback and/or mitigation plans in place in case of triggers from the system's monitoring and review regime?

8.14 Disclaimer

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